Dubai Metro, UAE
Rashidiya Depot

Structural Design Calculation
(Enhanced Design)
Building 12 – Traction Power Substation

July 2007

ATKINS
WS Atkins & Partners Overseas
Supported by Meinhardt Infrastructure

Client: **JT Metro JV**	**Contract No. (if any):**
Project Title: **Dubai Metro, UAE**	**Project No.:** **JRD4561**
Document No.: **DM001-E-MAM-CDP-DR-DCC-371949**	**Controlled Copy No.:**
Document Title: **Main Depot- Rashidiya** **Structural Design Calculation** **Building 12 – Traction Power Substation**	
Covering Letter / Transmittal Ref. No.: **To be advised**	**Date of Issue:** **21 Jul 2007**

Revision, Review and Approval/Authorization Records

A4	Issued for Approval	CK Lam Jul 07	P.Shepherd-Smith Jul 07	B. Maney Jul 07
A3	Issued for Approval	CK Lam Apr 06	P.Shepherd-Smith Apr 06	B. Maney Apr 06
A2	Issued for Approval	CK Lam Nov 06	P.Shepherd-Smith Nov 06	B. Maney Nov 06
A1	Internal Review	CK Lam Oct 06	P.Shepherd-Smith Oct 06	B. Maney Oct 06
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

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1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (P. Shepherd-Smith)
4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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1 INTRODUCTION

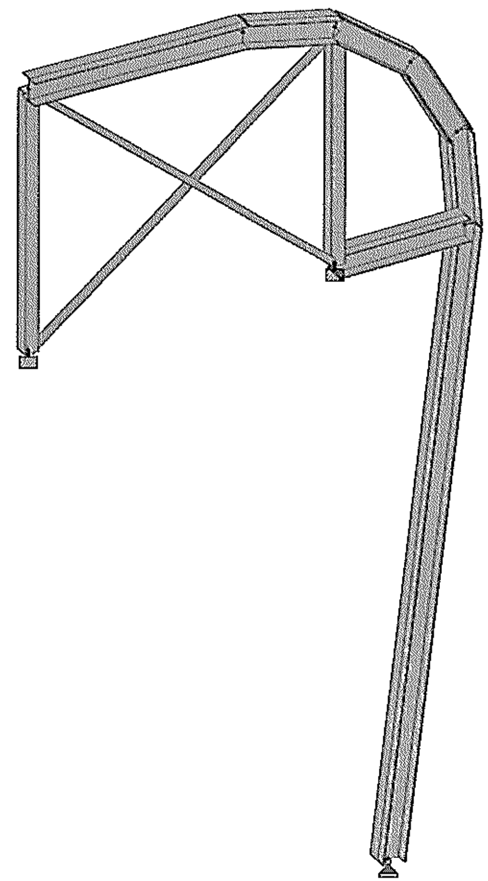
INTRODUCTION

This calculation report is submitted as a supplementary calculation to cover all new additions, amendments and updates to previously submitted report.

Please refer to Document no. DM001-E-MAM-CDP-DR-DCC-371949 Rev. A3 for previously submitted calculations and details.

2 ROOF SIDE FRAME ANALYSIS AND DESIGN

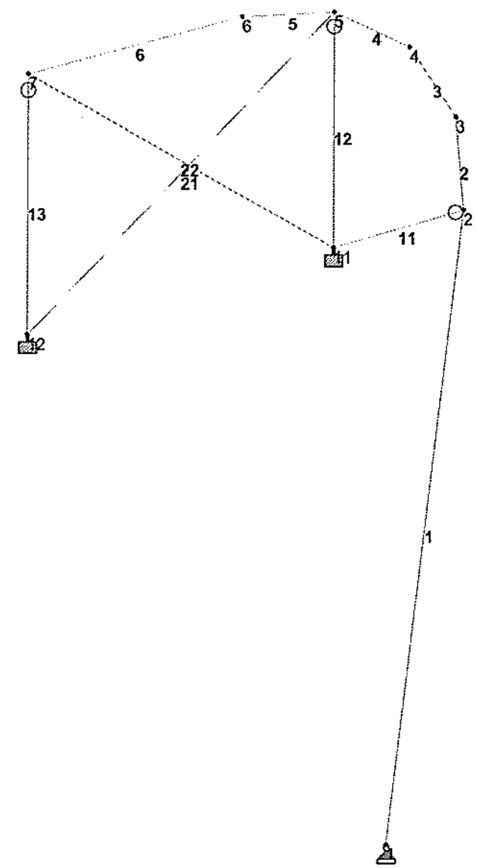
Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:29	



TPS BUILDING - SIDE FRAME

Load 1

Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



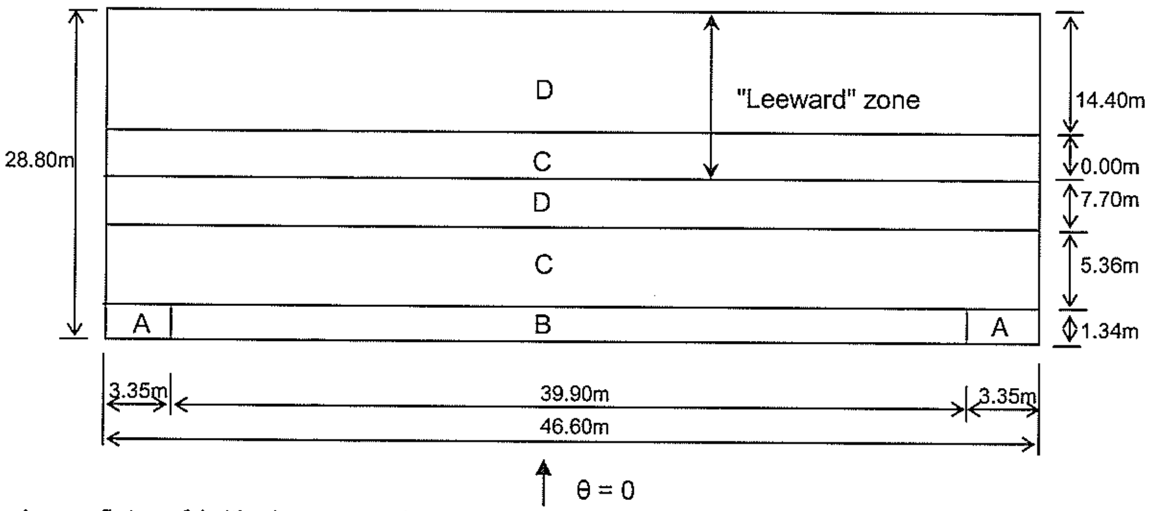
JOINT & MEMBER INCIDENCES

Load 1

Site:	Dubai Metro Project	Date:	9-Jul-07	Job No.:	68010.01
	TPS Building				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

PLAN



Design as flat roof (with sharp eaves)

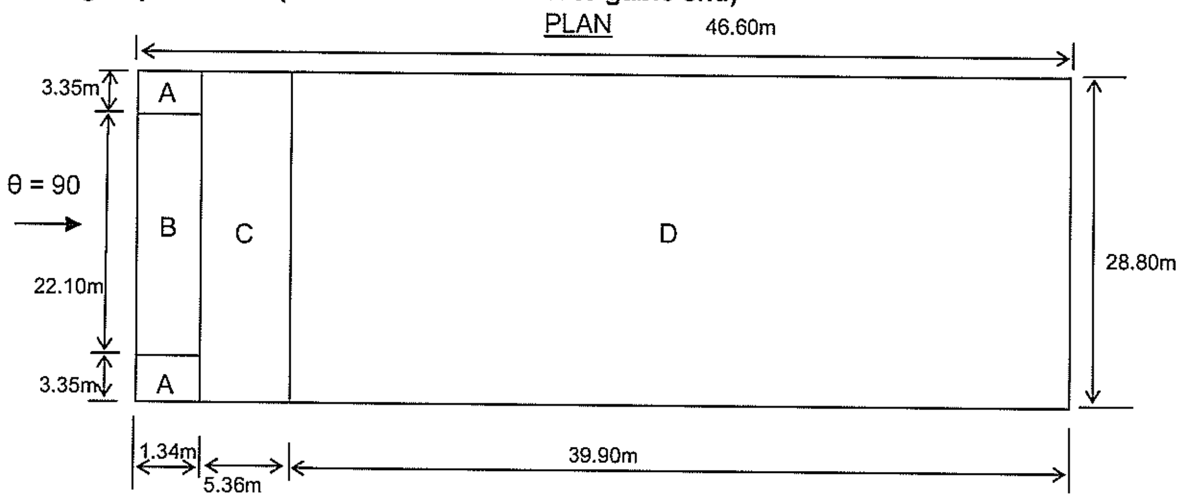
B = 46.6 m
H = 6.7 m
b = smaller of (B, 2H)
= 2 x 6.7
= 13.4 m

Zone	Windward Area (m^2)	Windward Cpe	Leeward Area	Leeward Cpe1	Leeward Cpe2
A	9.0	-2			
B	53.5	-1.4			
C	249.8	-0.7	0.0	-0.7	-0.7
D	358.8	-0.2	671.0	-0.2	0.2
Total	671.0		671.0		
Average		-0.51		-0.20	0.20

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers
MEINHARDT INFRASTRUCTURE

Site:	Dubai Metro Project TPS Building Wind Load Computation	Date:	9-Jul-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

B = 28.8 m
H = 6.7 m
b = smaller of (B, 2H)
= 2 x 6.7
= 13.4 m

For maximum uplift, assume zone C coefficient => Cpe = -0.7
For maximum downward force, assume zone D coefficient => Cpe = 0.2

Wind Load Computation

D = 28.80m m
H = 6.7 m
D/H = 4.30

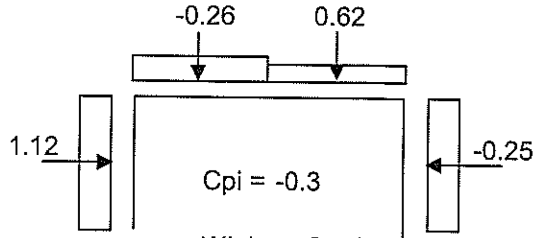
Elements	Effective Wind Speed Ve m/s	Dynamic Wind Pressure q kN/m^2	External Pressure Coefficient, Cpe				
			θ = 0 (normal to eave)			θ = 90 (normal to gable end)	
			Windward	Leeward			
Roof	45	1.24	-0.51	-0.20	0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.60	-0.5	-0.5	-0.8 (Zone B)	-0.8 (Zone B)

Calculation/Sketch

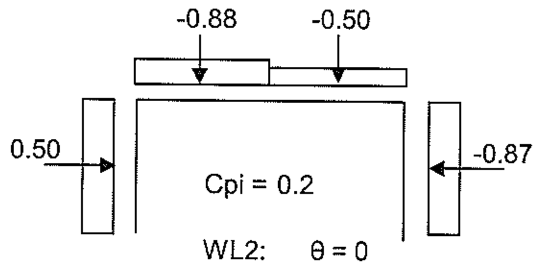
Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site:	Dubai Metro Project TPS Building Wind Load Computation	Date:	9-Jul-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

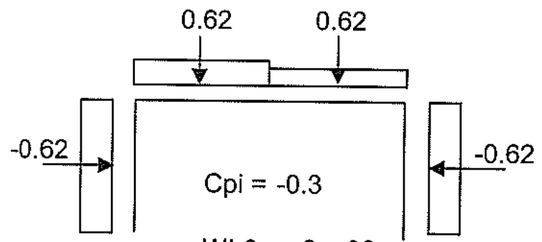
Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



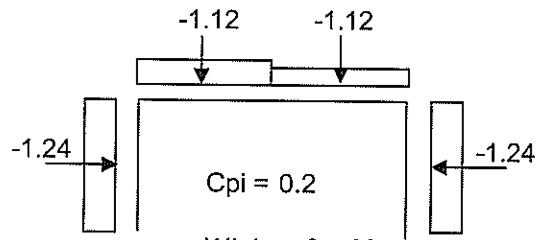
WL1: θ = 0
Internal suction
Roof Cpe for Leeward = 0.20



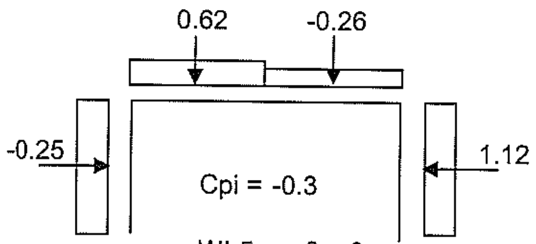
WL2: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20



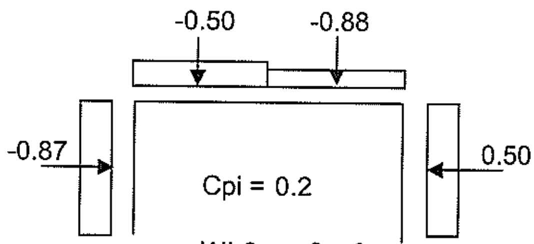
WL3: θ = 90
Internal suction
Roof Cpe = 0.20



WL4: θ = 90
Internal pressure
Roof Cpe = -0.70



WL5: θ = 0
Internal suction
Roof Cpe for Leeward = 0.20
(Reverse wind direction)



WL6: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20
(Reverse wind direction)

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	Part		
Job Title	Ref		
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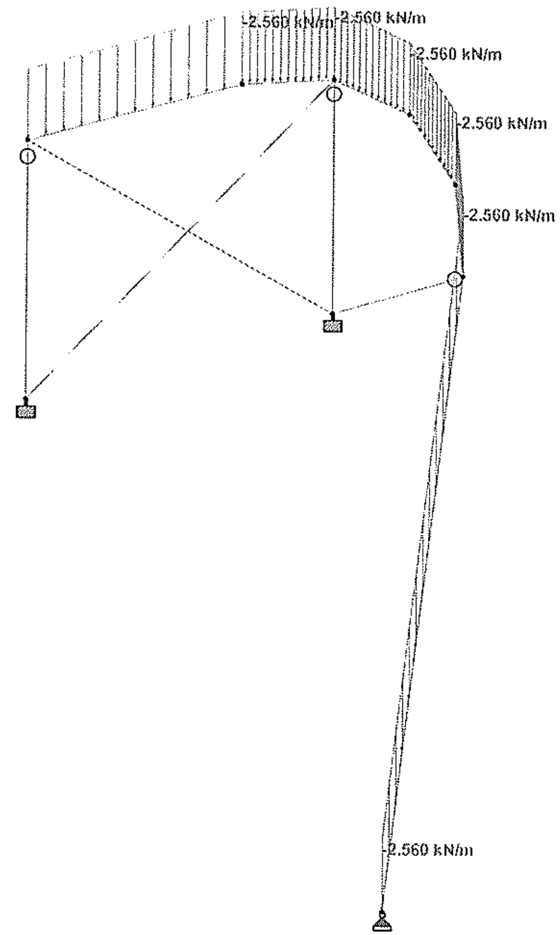
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



Load 2

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	Part		
Job Title	Ref		
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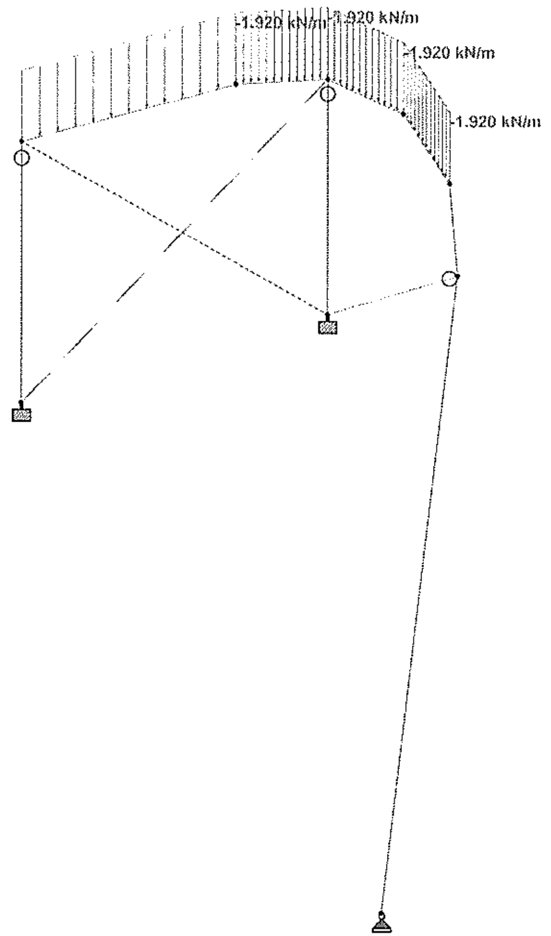
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



Load 3

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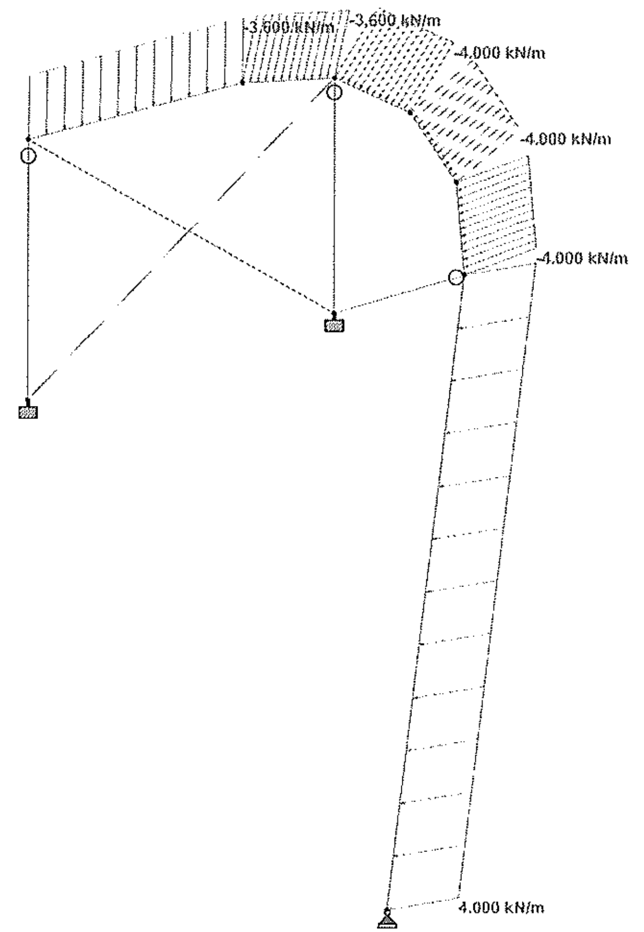
Job Title	Job No	Sheet No 1	Rev
	Part		
	Ref		
	By	Date JULY 2007	Chd
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



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Job Title	Ref		
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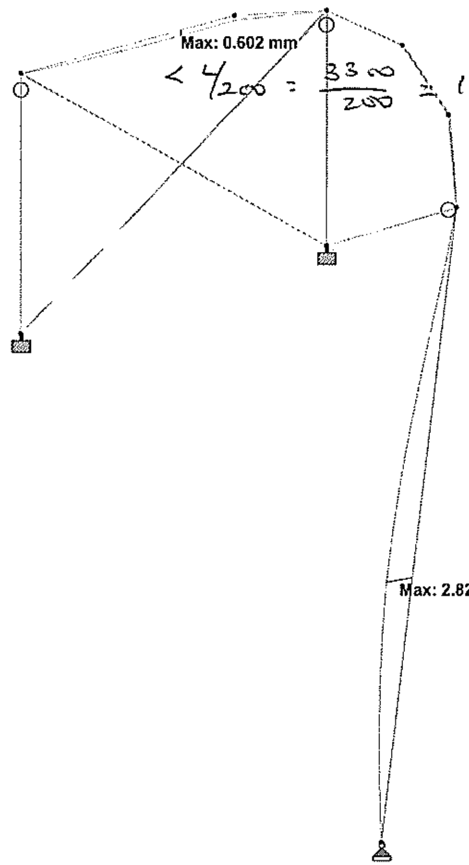
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



$< L/200 = \frac{3300}{200} = 16.5$ mm ok

$< L/200 = \frac{6700}{200} = 33.5$ mm ok

Load 2011 : Displacement
Displacement - mm

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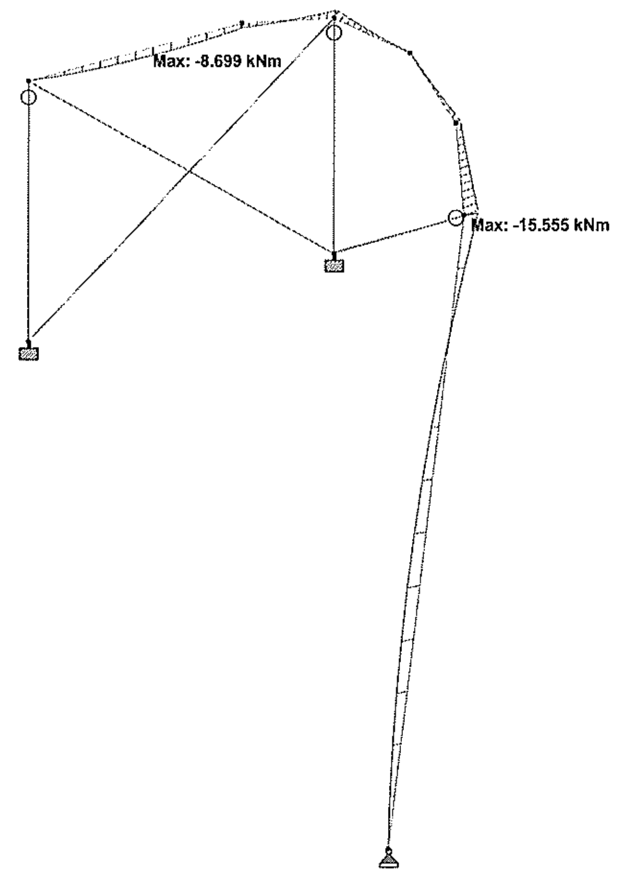
	Part			
Job Title	Ref			
	By	Date JULY 2007	Chd	
Client	File Steel_Roof-TPS.std		Date/Time 09-Jul-2007 16:37	



Load 2011 : Bending Z
Moment - kNm

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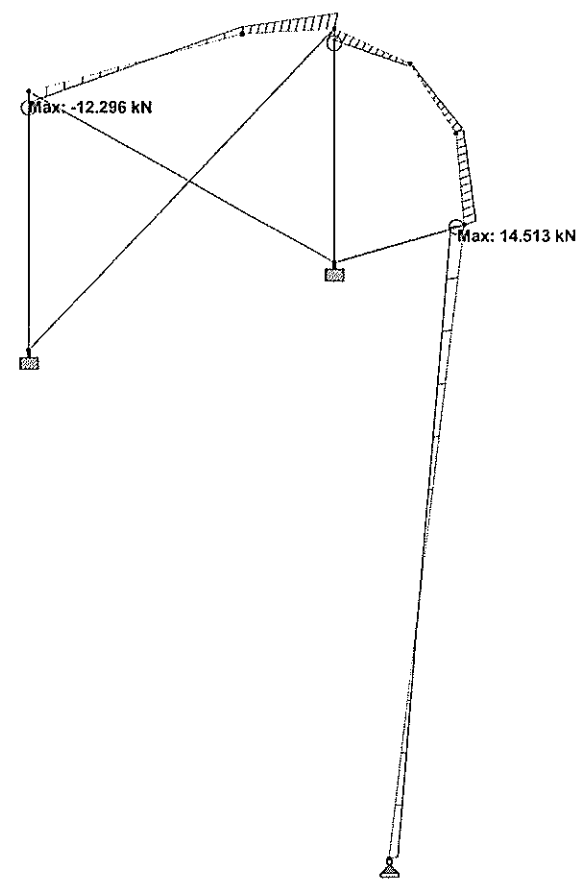
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37	



Load 2011 : Shear Y
Force - kN

STAAD.Pro for Windows Release 2006

```
 ****************************************************
 *                                                  *
 *           STAAD.Pro                              *
 *           Version  2006     Bld 1005.US          *
 *           Proprietary Program of                 *
 *           Research Engineers,  Intl.             *
 *           Date=    JUL  9, 2007                  *
 *           Time=    16:39:26                      *
 *                                                  *
 *      USER ID: Meinhardt Infrastructure Pte Ltd   *
 ****************************************************
```

```
     1. STAAD PLANE
INPUT FILE: Steel_Roof-TPS.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE JULY 2007
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1  19.450  0.000
     9. 2  20.273  6.705
    10. 3  20.183  7.739
    11. 4  19.679  8.647
    12. 5  18.850  9.272
    13. 6  17.839  9.506
    14. 7  15.501  9.547
    15. 11 18.850  6.705
    16. 12 15.501  6.705
    18. MEMBER INCIDENCES
    19. 1  1 2 6
    20. 11 2 11
    21. 12 11 5
    22. 13 12 7
    23. 21 12 5
    24. 22 7 11
    26. DEFINE MATERIAL START
    27. ISOTROPIC STEEL
    28. E 2.05E+008
    29. POISSON 0.3
    30. DENSITY 76.8195
    31. ALPHA 1.2E-005
    32. DAMP 0.03
    33. END DEFINE MATERIAL
    35. MEMBER PROPERTY BRITISH
    36. 1 TO 6 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    37. 11 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    38. 12 13 TABLE ST TUB1501506.3
    39. 21 22 TABLE ST UA90X90X6
    41. CONSTANTS
    42. MATERIAL STEEL ALL
    43. MEMBER TENSION
    44. 21 22
```



```
    46. MEMBER RELEASES
    47. 11 START MX MY MZ
    48. 12 13 END MX MY MZ
    50. SUPPORTS
    51. 1 PINNED
    52. 11 12 FIXED
    54. LOAD 1 SELFWEIGHT
    55. SELFWEIGHT Y -1
    57. LOAD 2 DEAD LOAD
    58. MEMBER LOAD
    59. *ASSUME ROOF M&E SERVICES LOAD = 0.3 KPA
    60. *ASSUME ROOF INSULATION AND ROOF SHEETING AND MEMBERS LOAD = 0.5 KPA
    61. 1 TO 6 UNI GY -2.56
    63. LOAD 3 LIVE LOAD
    64. *LL = 0.6 KPA
    65. MEMBER LOAD
    66. 3 TO 6 UNI GY -1.92
    68. LOAD 11 WL
    69. MEMBER LOAD
    70. 1 UNI Y 4
    71. 2 TO 4 UNI Y -4.0
    72. 5 6 UNI Y -3.6
    74. LOAD COMB 1001 1.4DL + 1.6LL
    75. 1 1.4 2 1.4 3 1.6
    76. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL
```



```
    77. 1 1.2 2 1.2 3 1.2 11 1.2
    79. LOAD COMB 2001 1.0DL + 1.0LL
    80. 1 1.0 2 1.0 3 1.0
    81. LOAD COMB 2011 DL + LL + WL
    82. 1 1.0 2 1.0 3 1.0 11 1.0
    84. PERFORM ANALYSIS
```

```
       P R O B L E M   S T A T I S T I C S
       -----------------------------------

 NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =      9/    11/     3
 ORIGINAL/FINAL BAND-WIDTH=      6/      3/      9 DOF
 TOTAL PRIMARY LOAD CASES =      4, TOTAL DEGREES OF FREEDOM =      19
 SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
 REQRD/AVAIL. DISK SPACE  =      12.0/ 366772.7 MB

**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=    1


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    2
```



```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=    3


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=   11


    85. LOAD LIST ALL
    86. PRINT SUPPORT REACTION
SUPPORT  REACTION
```


SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	0.41	5.03	0.00	0.00	0.00	0.00
	2	1.59	20.18	0.00	0.00	0.00	0.00
	3	0.19	1.49	0.00	0.00	0.00	0.00
	11	10.48	-4.79	0.00	0.00	0.00	0.00
	1001	3.11	37.69	0.00	0.00	0.00	0.00
	1011	15.21	26.30	0.00	0.00	0.00	0.00
	2001	2.19	26.70	0.00	0.00	0.00	0.00
	2011	12.67	21.92	0.00	0.00	0.00	0.00
11	1	-0.41	3.91	0.00	0.00	0.00	0.43
	2	-1.60	9.79	0.00	0.00	0.00	1.17
	3	-0.20	5.96	0.00	0.00	0.00	0.15
	11	29.67	15.50	0.00	0.00	0.00	-0.31
	1001	-3.14	28.71	0.00	0.00	0.00	2.48
	1011	32.94	42.18	0.00	0.00	0.00	1.73
	2001	-2.21	19.66	0.00	0.00	0.00	1.75
	2011	27.45	35.15	0.00	0.00	0.00	1.44
12	1	0.00	2.05	0.00	0.00	0.00	-0.01
	2	0.01	3.94	0.00	0.00	0.00	-0.03
	3	0.01	3.02	0.00	0.00	0.00	-0.02
	11	-2.07	3.75	0.00	0.00	0.00	0.06
	1001	0.03	13.21	0.00	0.00	0.00	-0.09
	1011	-2.46	15.31	0.00	0.00	0.00	0.00
	2001	0.02	9.01	0.00	0.00	0.00	-0.06
	2011	-2.05	12.75	0.00	0.00	0.00	0.00

```
 ************** END OF LATEST ANALYSIS RESULT **************

   87. PRINT FORCE ENVELOPE NSECTION 5 ALL
 FORCE    ENVELOPE NSECTION 5
   STAAD PLANE                                            -- PAGE NO.    5

       MEMBER FORCE ENVELOPE
       ---------------------

       ALL UNITS ARE KN   METE

 MEMB  DISTANCE          FY    LD        MZ    LD        FZ    LD        MY    LD

   1     0.00  MAX     1.51 1001      0.00 2011      0.00 2011      0.00 2011
               MIN   -11.89 1011      0.00 2011      0.00 2011      0.00 2011
         1.35  MAX     0.78 1001     12.10 1011      0.00 2011      0.00 2011
               MIN    -6.03 1011     -1.54 1001      0.00 2011      0.00 2011
         2.70  MAX     0.05 1001     16.29 1011      0.00 2011      0.00 2011
               MIN    -0.18   11     -2.10 1001      0.00 2011      0.00 2011
         4.05  MAX     5.69 1011     12.56 1011      0.00 2011      0.00 2011
               MIN    -0.68 1001     -1.68 1001      0.00 2011      0.00 2011
         5.40  MAX    11.55 1011      0.95   11      0.00 2011      0.00 2011
               MIN    -1.41 1001     -0.27 1001      0.00 2011      0.00 2011
         6.76  MAX    17.42 1011      2.12 1001      0.00 2011      0.00 2011
               MIN    -2.13 1001    -18.67 1011      0.00 2011      0.00 2011

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        1, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
 |
 | MAX.     17.42   6.76 1011        16.29    2.70 1011
 |           0.00   0.00    1         0.00    0.00    1      37.78 C    0.00 1001
 | MIN.    -11.89   0.00 1011       -18.67    6.76 1011
 |           0.00   6.76 2011         0.00    6.76 2011       3.48 T    6.76   11
 |-----------------------------------------------------------------------------

   2     0.00  MAX    15.12 1011     18.67 1011      0.00 2011      0.00 2011
               MIN    -0.84 1001     -2.12 1001      0.00 2011      0.00 2011
         0.21  MAX    14.06 1011     15.64 1011      0.00 2011      0.00 2011
               MIN    -0.92 1001     -1.93 1001      0.00 2011      0.00 2011
         0.42  MAX    12.99 1011     12.83 1011      0.00 2011      0.00 2011
               MIN    -1.00 1001     -1.73 1001      0.00 2011      0.00 2011
         0.62  MAX    11.93 1011     10.24 1011      0.00 2011      0.00 2011
               MIN    -1.08 1001     -1.52 1001      0.00 2011      0.00 2011
         0.83  MAX    10.86 1011      7.88 1011      0.00 2011      0.00 2011
               MIN    -1.16 1001     -1.28 1001      0.00 2011      0.00 2011
         1.04  MAX     9.80 1011      5.74 1011      0.00 2011      0.00 2011
               MIN    -1.24 1001     -1.03 1001      0.00 2011      0.00 2011
   STAAD PLANE                                            -- PAGE NO.    6

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        2, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
 |
 | MAX.     15.12   0.00 1011        18.67    0.00 1011
 |           0.00   0.00    1         0.00    0.00    1       9.09 C    0.00 1001
 | MIN.     -1.24   1.04 1001        -2.12    0.00 1001
 |           0.00   1.04 2011         0.00    1.04 2011       2.95 T    1.04   11
 |-----------------------------------------------------------------------------

   3     0.00  MAX     8.96 1011      5.74 1011      0.00 2011      0.00 2011
               MIN     0.00    2     -1.03 1001      0.00 2011      0.00 2011
         0.21  MAX     7.35 1011      4.18   11      0.00 2011      0.00 2011
               MIN    -0.26    2     -1.10 1001      0.00 2011      0.00 2011
         0.42  MAX     5.74 1011      2.97   11      0.00 2011      0.00 2011
               MIN    -0.81 1001     -1.01 1001      0.00 2011      0.00 2011
         0.62  MAX     4.54   11      1.95   11      0.00 2011      0.00 2011
               MIN    -1.56 1001     -0.77 1001      0.00 2011      0.00 2011
         0.83  MAX     3.71   11      1.09   11      0.00 2011      0.00 2011
               MIN    -2.32 1001     -0.37    2      0.00 2011      0.00 2011
         1.04  MAX     2.87   11      0.61 1011      0.00 2011      0.00 2011
               MIN    -3.07 1001     -0.13    2      0.00 2011      0.00 2011
```



```
 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        3, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
 |
 | MAX.      8.96   0.00 1011         5.74    0.00 1011
 |           0.00   0.00    1         0.00    0.00    1       4.63 C    0.00 1001
 | MIN.     -3.07   1.04 1001        -1.10    0.21 1001
 |           0.00   1.04 2011         0.00    1.04 2011       9.49 T    1.04 1011
 |-----------------------------------------------------------------------------

   4     0.00  MAX     0.04   11      0.61 1011      0.00 2011      0.00 2011
               MIN    -3.69 1001     -0.13    2      0.00 2011      0.00 2011
         0.21  MAX    -0.48    1      1.45 1011      0.00 2011      0.00 2011
               MIN    -5.03 1011      0.06    1      0.00 2011      0.00 2011
         0.42  MAX    -0.58    1      2.70 1011      0.00 2011      0.00 2011
               MIN    -7.04 1011      0.17    1      0.00 2011      0.00 2011
         0.62  MAX    -0.68    1      4.37 1011      0.00 2011      0.00 2011
               MIN    -9.05 1011      0.30    1      0.00 2011      0.00 2011
         0.83  MAX    -0.78    1      6.45 1011      0.00 2011      0.00 2011
               MIN   -11.05 1011      0.45    1      0.00 2011      0.00 2011
         1.04  MAX    -0.88    1      8.96 1011      0.00 2011      0.00 2011
               MIN   -13.06 1011      0.62    1      0.00 2011      0.00 2011
   STAAD PLANE                                            -- PAGE NO.    7

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        4, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
 |
 | MAX.      0.04   0.00   11         8.96    1.04 1011
 |           0.00   0.00    1         0.00    0.00    1       0.07 C    0.00    3
 | MIN.    -13.06   1.04 1011        -0.13    0.00    2
 |           0.00   1.04 2011         0.00    1.04 2011      12.85 T    1.04 1011
 |-----------------------------------------------------------------------------

   5     0.00  MAX    19.77 1011      8.96 1011      0.00 2011      0.00 2011
               MIN     1.15    1      0.62    1      0.00 2011      0.00 2011
         0.21  MAX    17.64 1011      5.08 1011      0.00 2011      0.00 2011
               MIN     1.03    1      0.40    1      0.00 2011      0.00 2011
         0.42  MAX    15.51 1011      1.96 1001      0.00 2011      0.00 2011
               MIN     0.91    1      0.01   11      0.00 2011      0.00 2011
         0.62  MAX    13.38 1011      0.08    2      0.00 2011      0.00 2011
               MIN     0.79    1     -1.36 1011      0.00 2011      0.00 2011
         0.83  MAX    11.25 1011     -0.13    1      0.00 2011      0.00 2011
               MIN     0.67    1     -3.92 1011      0.00 2011      0.00 2011
         1.04  MAX     9.12 1011     -0.26    1      0.00 2011      0.00 2011
               MIN     0.55    1     -6.03 1011      0.00 2011      0.00 2011

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        5, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
 |
 | MAX.     19.77   0.00 1011         8.96    0.00 1011
 |           0.00   0.00    1         0.00    0.00    1       4.96 C    0.00 1001
 | MIN.      0.55   1.04    1        -6.03    1.04 1011
 |           0.00   1.04 2011         0.00    1.04 2011       0.26 C    1.04    1
 |-----------------------------------------------------------------------------

   6     0.00  MAX     9.60 1011     -0.26    1      0.00 2011      0.00 2011
               MIN     0.59    1     -6.03 1011      0.00 2011      0.00 2011
         0.47  MAX     4.73 1011     -0.47    1      0.00 2011      0.00 2011
               MIN     0.31    1     -9.38 1011      0.00 2011      0.00 2011
         0.94  MAX     0.20 1001     -0.55    1      0.00 2011      0.00 2011
               MIN    -0.27   11    -10.45 1011      0.00 2011      0.00 2011
         1.40  MAX    -0.25    1     -0.50    1      0.00 2011      0.00 2011
               MIN    -5.01 1011     -9.25 1011      0.00 2011      0.00 2011
         1.87  MAX    -0.53    1     -0.31    1      0.00 2011      0.00 2011
               MIN    -9.88 1011     -5.76 1011      0.00 2011      0.00 2011
         2.34  MAX    -0.81    1      0.00 2011      0.00 2011      0.00 2011
               MIN   -14.76 1011      0.00 2011      0.00 2011      0.00 2011
   STAAD PLANE                                            -- PAGE NO.    8

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB        6, AMONGST ALL SECT LOCATIONS
 |            FY/   DIST   LD           MZ/   DIST   LD
 |            FZ    DIST   LD           MY    DIST   LD        FX         DIST  LD
```

```
| MAX.       9.60   0.00 1011        0.00   2.34    1
|            0.00   0.00    1        0.00   0.00    1     1.74 C    0.00 1001
| MIN.     -14.76   2.34 1011      -10.45   0.94 1011
|            0.00   2.34 2011        0.00   2.34 2011     0.11 T    2.34   11
|-------------------------------------------------------------------------------

 11     0.00  MAX     0.30   11       0.00 2011       0.00 2011       0.00 2011
              MIN    -1.18 1001       0.00 2011       0.00 2011       0.00 2011
        0.28  MAX     0.30   11       0.37 1001       0.00 2011       0.00 2011
              MIN    -1.42 1001      -0.09   11       0.00 2011       0.00 2011
        0.57  MAX     0.30   11       0.81 1001       0.00 2011       0.00 2011
              MIN    -1.65 1001      -0.17   11       0.00 2011       0.00 2011
        0.85  MAX     0.30   11       1.31 1001       0.00 2011       0.00 2011
              MIN    -1.89 1001      -0.26   11       0.00 2011       0.00 2011
        1.14  MAX     0.30   11       1.88 1001       0.00 2011       0.00 2011
              MIN    -2.13 1001      -0.34   11       0.00 2011       0.00 2011
        1.42  MAX     0.30   11       2.52 1001       0.00 2011       0.00 2011
              MIN    -2.37 1001      -0.43   11       0.00 2011       0.00 2011

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      11, AMONGST ALL SECT LOCATIONS
|           FY/   DIST   LD          MZ/   DIST   LD
|           FZ    DIST   LD          MY    DIST   LD          FX         DIST   LD
|
| MAX.      0.30   0.00   11         2.52   1.42 1001
|           0.00   0.00    1         0.00   0.00    1     31.70 C    0.00 1011
| MIN.     -2.37   1.42 1001        -0.43   1.42   11
|           0.00   1.42 2011         0.00   1.42 2011      4.73 T    1.42 1001
|-------------------------------------------------------------------------------

 12     0.00  MAX     0.05   11       0.12   11       0.00 2011       0.00 2011
              MIN    -0.02 1001      -0.04 1001       0.00 2011       0.00 2011
        0.51  MAX     0.05   11       0.10   11       0.00 2011       0.00 2011
              MIN    -0.02 1001      -0.03 1001       0.00 2011       0.00 2011
        1.03  MAX     0.05   11       0.07   11       0.00 2011       0.00 2011
              MIN    -0.02 1001      -0.02 1001       0.00 2011       0.00 2011
        1.54  MAX     0.05   11       0.05   11       0.00 2011       0.00 2011
              MIN    -0.02 1001      -0.02 1001       0.00 2011       0.00 2011
        2.05  MAX     0.05   11       0.02   11       0.00 2011       0.00 2011
              MIN    -0.02 1001      -0.01 1001       0.00 2011       0.00 2011
        2.57  MAX     0.05   11       0.00 2011       0.00 2011       0.00 2011
              MIN    -0.02 1001       0.00 2011       0.00 2011       0.00 2011
   STAAD PLANE                                                 -- PAGE NO.    9

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      12, AMONGST ALL SECT LOCATIONS
|           FY/   DIST   LD          MZ/   DIST   LD
|           FZ    DIST   LD          MY    DIST   LD          FX         DIST   LD
|
| MAX.      0.05   0.00   11         0.12   0.00   11
|           0.00   0.00    1         0.00   0.00    1     41.41 C    0.00 1011
| MIN.     -0.02   2.57 1001        -0.04   0.00 1001
|           0.00   2.57 2011         0.00   2.57 2011      2.39 C    2.57    1
|-------------------------------------------------------------------------------

 13     0.00  MAX     0.02   11       0.06   11       0.00 2011       0.00 2011
              MIN    -0.03 1001      -0.09 1001       0.00 2011       0.00 2011
        0.57  MAX     0.02   11       0.05   11       0.00 2011       0.00 2011
              MIN    -0.03 1001      -0.07 1001       0.00 2011       0.00 2011
        1.14  MAX     0.02   11       0.04   11       0.00 2011       0.00 2011
              MIN    -0.03 1001      -0.05 1001       0.00 2011       0.00 2011
        1.71  MAX     0.02   11       0.02   11       0.00 2011       0.00 2011
              MIN    -0.03 1001      -0.04 1001       0.00 2011       0.00 2011
        2.27  MAX     0.02   11       0.01   11       0.00 2011       0.00 2011
              MIN    -0.03 1001      -0.02 1001       0.00 2011       0.00 2011
        2.84  MAX     0.02   11       0.00 2011       0.00 2011       0.00 2011
              MIN    -0.03 1001       0.00 2011       0.00 2011       0.00 2011

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      13, AMONGST ALL SECT LOCATIONS
|           FY/   DIST   LD          MZ/   DIST   LD
|           FZ    DIST   LD          MY    DIST   LD          FX         DIST   LD
|
| MAX.      0.02   0.00   11         0.06   0.00   11
|           0.00   0.00    1         0.00   0.00    1     16.98 C    0.00 1011
| MIN.     -0.03   2.84 1001        -0.09   0.00 1001
|           0.00   2.84 2011         0.00   2.84 2011      1.09 C    2.84    1
|-------------------------------------------------------------------------------
```



```
 21     0.00  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        0.84  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        1.69  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        2.53  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        3.38  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        4.22  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
   STAAD PLANE                                                 -- PAGE NO.   10

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      21, AMONGST ALL SECT LOCATIONS
|           FY/   DIST   LD          MZ/   DIST   LD
|           FZ    DIST   LD          MY    DIST   LD          FX         DIST   LD
|
| MAX.      0.00   0.00    1         0.00   0.00    1
|           0.00   0.00    1         0.00   0.00    1      0.00      0.00    1
| MIN.      0.00   4.22 2011         0.00   4.22 2011
|           0.00   4.22 2011         0.00   4.22 2011      3.10 T    4.22 1011
|-------------------------------------------------------------------------------

 22     0.00  MAX     0.19 1001       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00   11       0.00 2011       0.00 2011       0.00 2011
        0.88  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        1.76  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        2.64  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        3.51  MAX     0.00 2011       0.00 2011       0.00 2011       0.00 2011
              MIN     0.00 2011       0.00 2011       0.00 2011       0.00 2011
        4.39  MAX     0.00   11       0.00 2011       0.00 2011       0.00 2011
              MIN    -0.19 1001       0.00 2011       0.00 2011       0.00 2011

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      22, AMONGST ALL SECT LOCATIONS
|           FY/   DIST   LD          MZ/   DIST   LD
|           FZ    DIST   LD          MY    DIST   LD          FX         DIST   LD
|
| MAX.      0.19   0.00 1001         0.00   0.00    1
|           0.00   0.00    1         0.00   0.00    1      0.00      0.00   11
| MIN.     -0.19   4.39 1001         0.00   4.39 2011
|           0.00   4.39 2011         0.00   4.39 2011      2.23 T    0.00 1001
|-------------------------------------------------------------------------------

 ********** END OF FORCE ENVELOPE FROM INTERNAL STORAGE **********

  88. LOAD LIST 2001 2011
  89. PRINT JOINT DISPL
 JOINT    DISPL
   STAAD PLANE                                                 -- PAGE NO.   11


 JOINT DISPLACEMENT (CM   RADIANS)    STRUCTURE TYPE = PLANE
 ------------------

 JOINT  LOAD   X-TRANS   Y-TRANS   Z-TRANS   X-ROTAN   Y-ROTAN   Z-ROTAN

     1 2001     0.0000    0.0000    0.0000    0.0000    0.0000   -0.0002
       2011     0.0000    0.0000    0.0000    0.0000    0.0000    0.0015
     2 2001     0.0003   -0.0069    0.0000    0.0000    0.0000    0.0001
       2011    -0.0023   -0.0051    0.0000    0.0000    0.0000   -0.0006
     3 2001    -0.0037   -0.0076    0.0000    0.0000    0.0000    0.0000
       2011     0.0184   -0.0034    0.0000    0.0000    0.0000    0.0000
     4 2001    -0.0029   -0.0071    0.0000    0.0000    0.0000    0.0000
       2011     0.0134   -0.0058    0.0000    0.0000    0.0000    0.0001
     5 2001    -0.0025   -0.0064    0.0000    0.0000    0.0000    0.0001
       2011     0.0080   -0.0119    0.0000    0.0000    0.0000    0.0002
     6 2001    -0.0071   -0.0270    0.0000    0.0000    0.0000    0.0001
       2011    -0.0008   -0.0512    0.0000    0.0000    0.0000    0.0003
     7 2001    -0.0065   -0.0033    0.0000    0.0000    0.0000   -0.0003
```

```
       2011    0.0001   -0.0053    0.0000    0.0000    0.0000   -0.0005
    11 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
       2011    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    12 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
       2011    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000

 ************** END OF LATEST ANALYSIS RESULT **************

   90. PARAMETER 1
   91. CODE BS5950
   92. PY 345000 ALL
   93. TRACK 2 ALL
   94. CHECK CODE ALL
 STEEL DESIGN

                       STAAD.Pro CODE CHECKING - (BSI )
                       ************************

  PROGRAM CODE REVISION V2.9_5950-1_2000
     STAAD PLANE                                          -- PAGE NO.   12

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

      1    TAPERED       PASS     BS-4.7   (C)        0.061       2011
                       14.47 C         0.00           13.57          2.81
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     1 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  1048.0   PT=     0.0    MB=  165.0   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      2    TAPERED       PASS     BS-4.7   (C)        0.059       2011
                        3.30 C         0.00           15.56          0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     2 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.2   PT=     0.0    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      3    TAPERED       PASS     BS-4.8.2.2          0.019       2011
                        3.29 T         0.00            4.78          0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     3 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.1   PT=  2693.8    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      4    TAPERED       PASS     BS-4.8.2.2          0.032       2011
                       10.71 T         0.00           -7.46          1.04
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     4 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC= 11908.2   PT=  2693.8    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      5    TAPERED       PASS     BS-4.2.3-(Y)        0.040       2011
                        3.90 C         0.00            7.46          0.00
     STAAD PLANE                                          -- PAGE NO.   13

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================
```



```
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     5 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.3   PT=     0.0    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      6    TAPERED       PASS     BS-4.7   (C)        0.033       2011
                        0.98 C         0.00           -8.70          0.97
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     6 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2346.9   PT=     0.0    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
     11    TAPERED       PASS     BS-4.7   (C)        0.015       2011
                       26.41 C         0.00           -1.35          1.42
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    11 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2583.3   PT=  2693.8    MB=  270.6   |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2  |
 |  d   = 200 mm   :   t  =   8 mm          : a  =    0 mm : pyf = 344 N/mm2  |
 |  BS-4.4.3.2       status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
     STAAD PLANE                                          -- PAGE NO.   14

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX            MY             MZ       LOCATION
 =======================================================================

     12 ST TUB1501506.3  PASS     BS-4.8.3.3.3        0.032       2011
                       34.51 C         0.00            0.09          -
 =======================================================================
   MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
     Member Length =     256.70
     Gross Area =    35.80  Net Area =    35.80  Eff. Area =    35.80

                                  z-z  axis        y-y  axis
     Moment of inertia       :    1223.000         1223.000
     Plastic modulus         :     192.000          192.000
     Elastic modulus         :     163.067          163.067
     Effective modulus       :     163.067          163.067
     Shear Area              :      17.900           17.900

   DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class           :   SEMI-COMPACT
     Squash Load             :   1235.10
     Axial force/Squash load :     0.027

                                  z-z  axis        y-y  axis
     Compression Capacity    :      1141.3           1141.3
     Moment Capacity         :        66.2             66.2
     Reduced Moment Capacity :        66.2             66.2
     Shear Capacity          :       370.5            370.5

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                                  x-x axis         y-y axis
     Slenderness             :      43.919           43.919
     Radius of gyration (cm) :       5.845            5.845
     Effective Length        :       2.567            2.567

     LTB check unnecessary for this section

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE         RATIO   LOAD     FX       VY       VZ       MZ       MY
   BS-4.7   (C)    0.030   2011     34.5      -        -        -        -
   BS-4.8.3.2      0.028   2011     33.9     0.0      0.0      0.0      0.0
```


018

```
 BS-4.8.3.3.1   0.032  2011    34.5       -        -       0.1      0.0
 BS-4.8.3.3.3   0.032  2011    34.5       -        -       0.1      0.0
 ANNEX I.1      0.001  2011    34.5       -        -       0.1      0.0
  Torsion and deflections have not been considered in the design.
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/     RATIO/      LOADING/
                            FX          MY               MZ        LOCATION
=======================================================================

     13 ST TUB1501506.3  PASS      BS-4.8.3.3.3       0.013       2011
                        14.15 C        0.00            0.00          -
=======================================================================
  MATERIAL DATA
     Grade of steel          =  S 275
     Modulus of elasticity   =  205 kN/mm2
     Design Strength  (py)   =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =     284.20
     Gross Area =   35.80  Net Area =   35.80  Eff. Area =   35.80

                              z-z  axis       y-y  axis
     Moment of inertia      :   1223.000        1223.000
     Plastic modulus        :    192.000         192.000
     Elastic modulus        :    163.067         163.067
     Effective modulus      :    163.067         163.067
     Shear Area             :     17.900          17.900

  DESIGN DATA (units - kN,m)  BS5950-1/2000
     Section Class          :  SEMI-COMPACT
     Squash Load            :  1235.10
     Axial force/Squash load :   0.011

                              z-z  axis       y-y  axis
     Compression Capacity   :     1117.9          1117.9
     Moment Capacity        :       66.2            66.2
     Reduced Moment Capacity :      66.2            66.2
     Shear Capacity         :      370.5           370.5

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                              x-x axis        y-y axis
     Slenderness            :     48.624          48.624
     Radius of gyration (cm) :     5.845           5.845
     Effective Length       :      2.842           2.842

     LTB check unnecessary for this section

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.7    (C)   0.013  2011    14.2       -        -        -        -
  BS-4.8.3.2      0.011  2011    14.2      0.0      0.0      0.0      0.0
  BS-4.8.3.3.1    0.013  2011    14.2       -        -       0.0      0.0
  BS-4.8.3.3.3    0.013  2011    14.2       -        -       0.0      0.0
   Torsion and deflections have not been considered in the design.
```



```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/     RATIO/      LOADING/
                            FX          MY               MZ        LOCATION
=======================================================================

     21 ST UA90X90X6       PASS      BS-4.6    (T)     0.007       2011
                          2.58 T        0.00           0.00         0.00
=======================================================================
  MATERIAL DATA
     Grade of steel          =  S 275
     Modulus of elasticity   =  205 kN/mm2
     Design Strength  (py)   =  344  N/mm2

  SECTION PROPERTIES (units - cm)
```

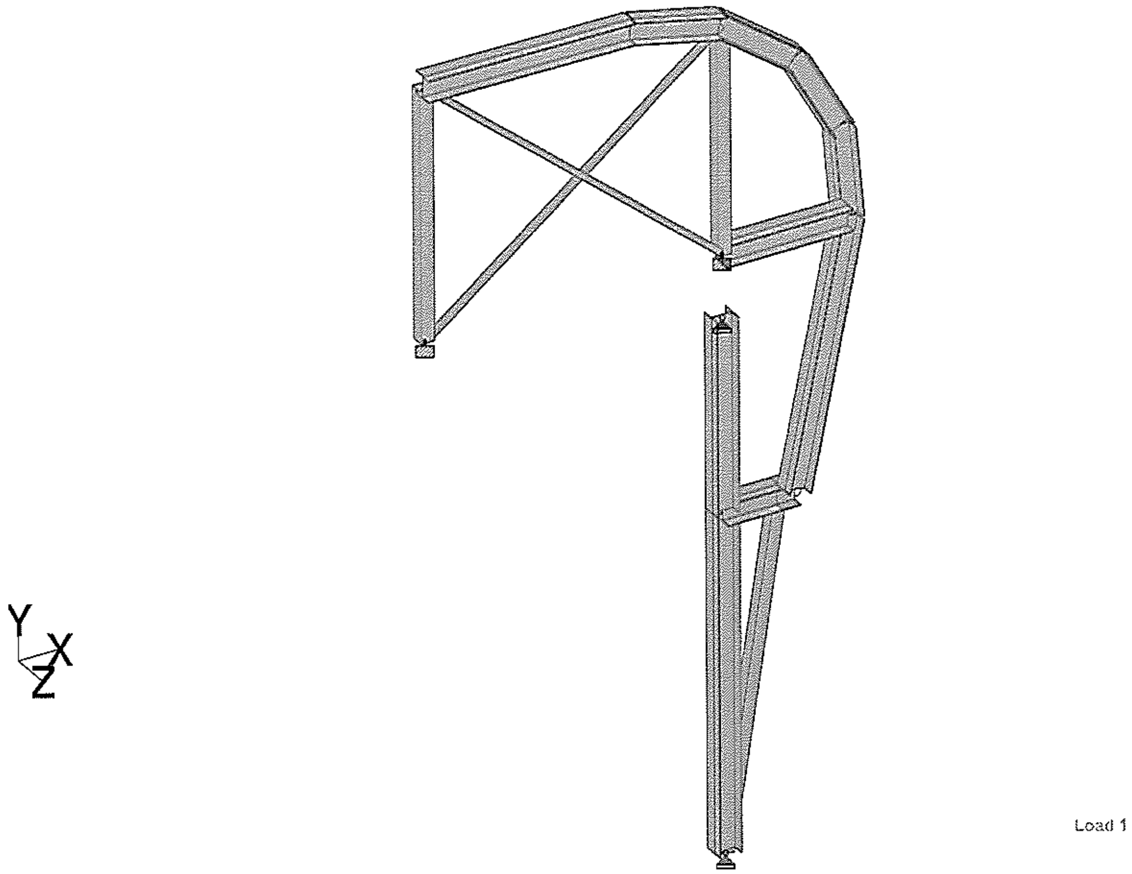


```
     Member Length =     421.96
     Gross Area =   10.60  Net Area =   10.60  Eff. Area =   10.60

                              z-z  axis       y-y  axis
     Moment of inertia      :     33.585         131.618
     Plastic modulus        :     17.554          37.227
     Elastic modulus        :      9.752          20.682
     Effective modulus      :      9.752          20.682
     Shear Area             :      3.240           3.240

  DESIGN DATA (units - kN,m)  BS5950-1/2000
     Section Class          :  SLENDER

                              z-z  axis       y-y  axis
     Tension Capacity       :      365.7           365.7
     Shear Capacity         :       67.1            67.1

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

     LTB Moment Capacity (kNm) and LTB Length (m):      4.09,     4.220
     LTB Coefficients & Associated Moments (kNm):
     mLT =    1.00  :   mx =    1.00   :  my =    1.00   :  myx =    1.00
     Mlt =     0.00 :   Mx =     0.00  :  My =    0.00   :   My =    0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.6    (T)   0.007  2011     2.6       -        -        -        -
   Torsion and deflections have not been considered in the design.
```



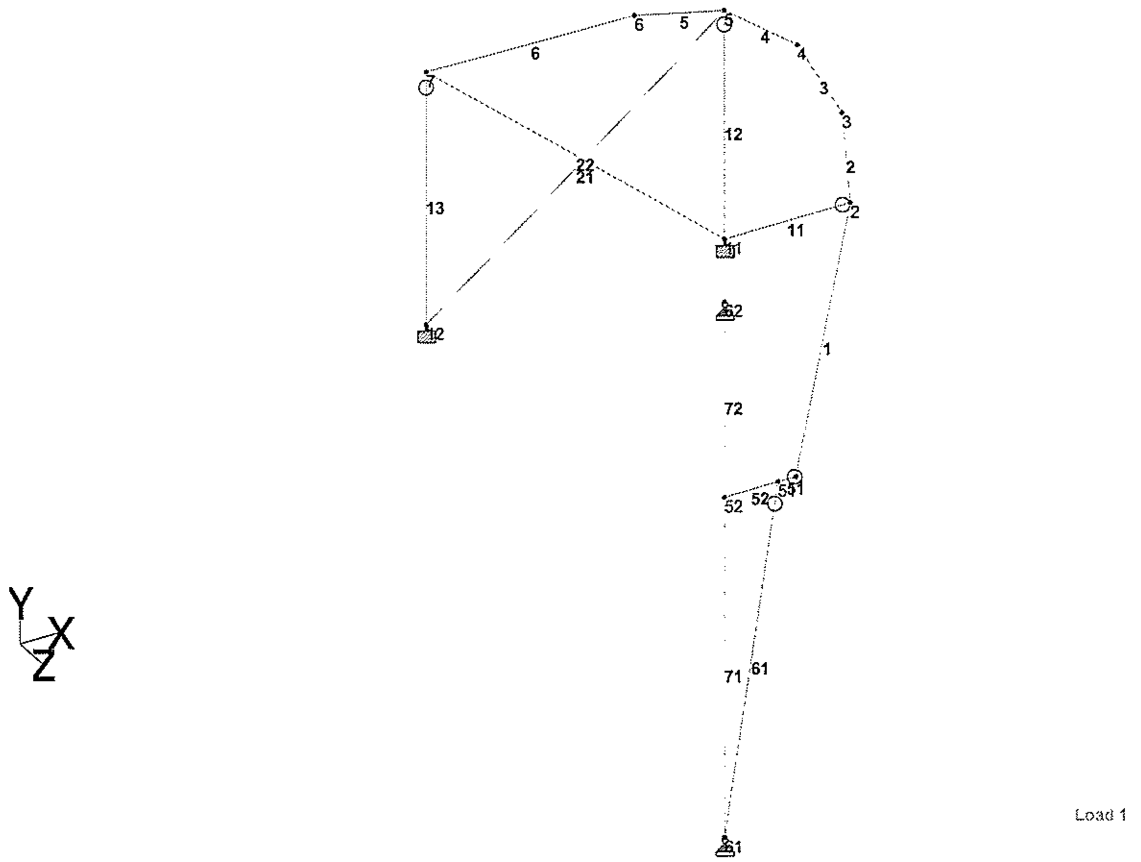
```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE          RESULT/   CRITICAL COND/     RATIO/      LOADING/
                            FX          MY               MZ        LOCATION
=======================================================================

     22 ST UA90X90X6       PASS      BS-4.6    (T)     0.004       2001
                          1.53 T        0.00           0.00         0.00
=======================================================================
  MATERIAL DATA
     Grade of steel          =  S 275
     Modulus of elasticity   =  205 kN/mm2
     Design Strength  (py)   =  344  N/mm2

  SECTION PROPERTIES (units - cm)
     Member Length =     439.24
     Gross Area =   10.60  Net Area =   10.60  Eff. Area =   10.60

                              z-z  axis       y-y  axis
     Moment of inertia      :     33.585         131.618
     Plastic modulus        :     17.554          37.227
     Elastic modulus        :      9.752          20.682
     Effective modulus      :      9.752          20.682
     Shear Area             :      3.240           3.240

  DESIGN DATA (units - kN,m)  BS5950-1/2000
     Section Class          :  SLENDER

                              z-z  axis       y-y  axis
     Tension Capacity       :      365.7           365.7
     Shear Capacity         :       67.1            67.1

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

     LTB Moment Capacity (kNm) and LTB Length (m):      3.99,     4.392
     LTB Coefficients & Associated Moments (kNm):
     mLT =    1.00  :   mx =    1.00   :  my =    1.00   :  myx =    1.00
     Mlt =     0.00 :   Mx =     0.00  :  My =    0.00   :   My =    0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.6    (T)   0.004  2001     1.5       -        -        -        -
   Torsion and deflections have not been considered in the design.

  ************** END OF TABULATED RESULT OF DESIGN **************
```

Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



SIDE FRAME-BRU PANEL AREA (TPS)

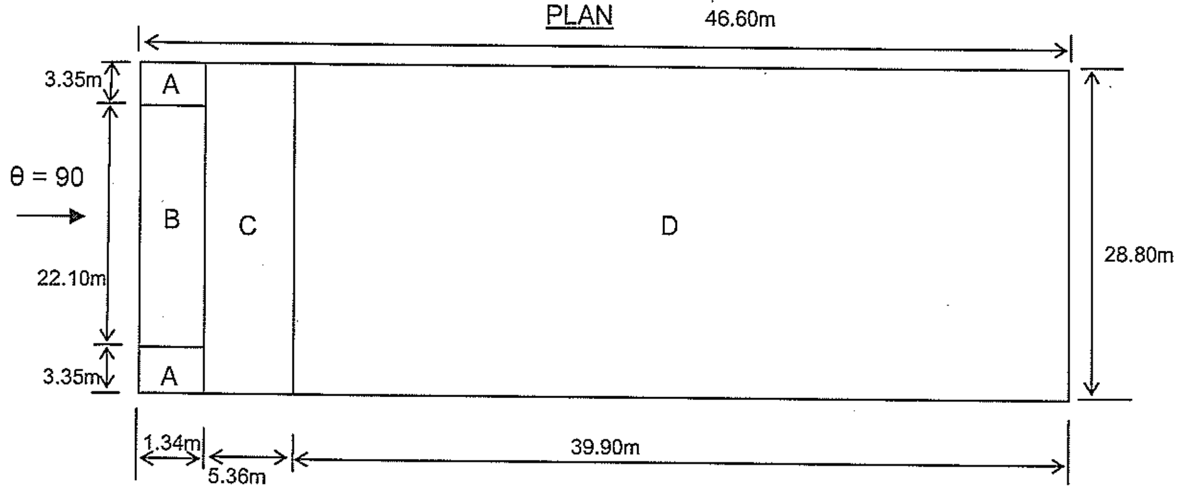
Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



JOINT & MEMBER INCIDENCES

Site:	Dubai Metro Project	Date:	9-Jul-07	Job No.:	68010.01
	TPS Building				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

B = 28.8 m
H = 6.7 m
b = smaller of (B, 2H)
= 2 x 6.7
= 13.4 m

For maximum uplift, assume zone C coefficient => Cpe = -0.7
For maximum downward force, assume zone D coefficient => Cpe = 0.2

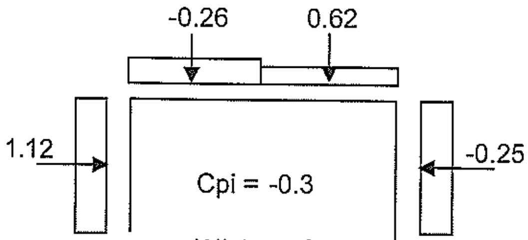
Wind Load Computation

D = 28.80m m
H = 6.7 m
D/H = 4.30

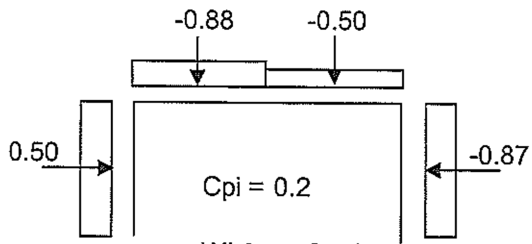
Elements	Effective Wind Speed Ve m/s	Dynamic Wind Pressure q kN/m^2	External Pressure Coefficient, Cpe				
			θ = 0 (normal to eave)			θ = 90 (normal to gable end)	
			Windward	Leeward			
Roof	45	1.24	-0.51	-0.20	0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.60	-0.5	-0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site:	Dubai Metro Project TPS Building Wind Load Computation	Date:	9-Jul-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

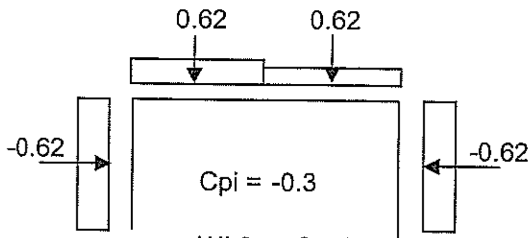
Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



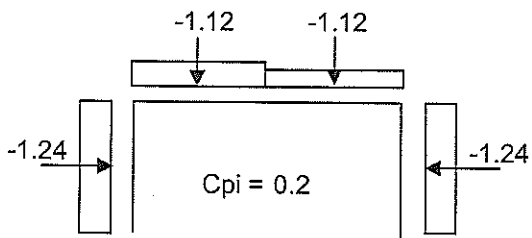
WL1: θ = 0
Internal suction
Roof Cpe for Leeward = 0.20



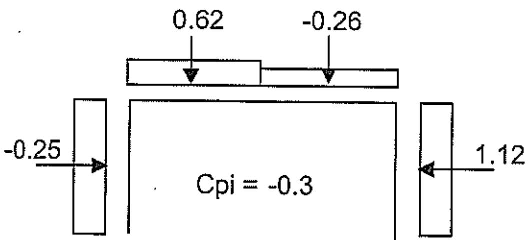
WL2: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20



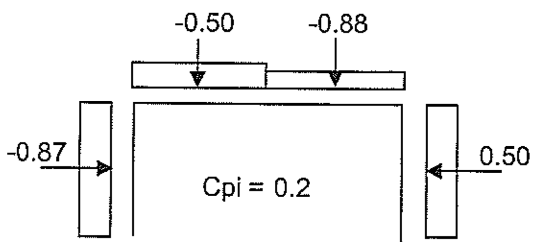
WL3: θ = 90
Internal suction
Roof Cpe = 0.20



WL4: θ = 90
Internal pressure
Roof Cpe = -0.70



WL5: θ = 0
Internal suction
Roof Cpe for Leeward = 0.20
(Reverse wind direction)



WL6: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20
(Reverse wind direction)

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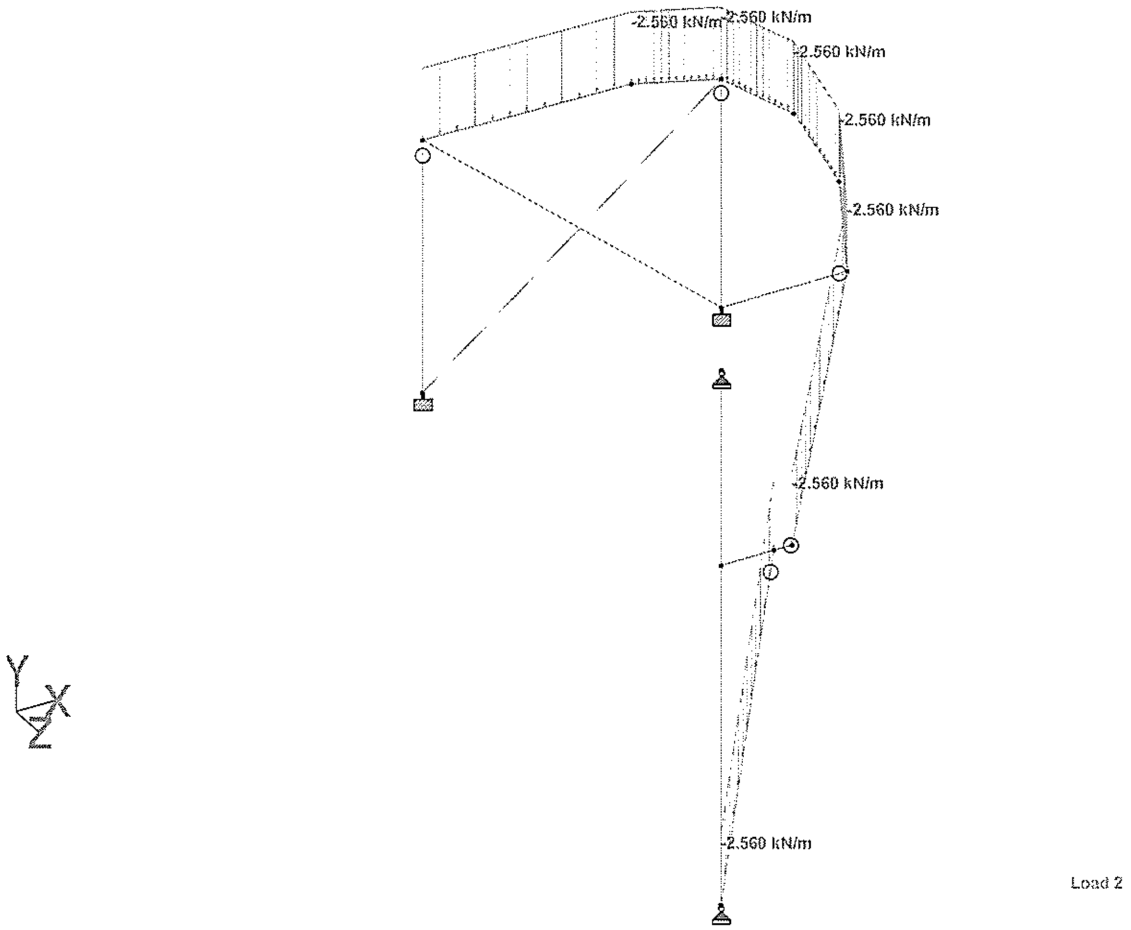
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



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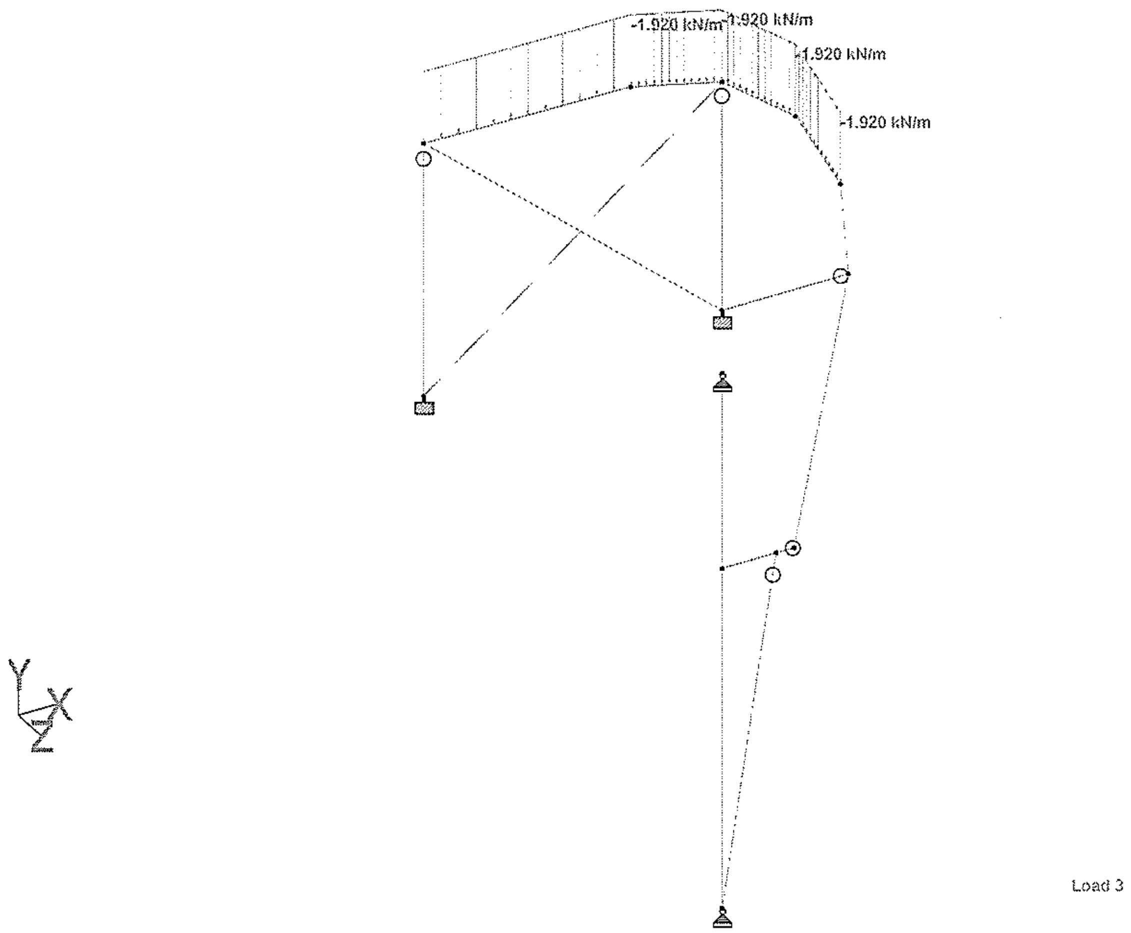
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



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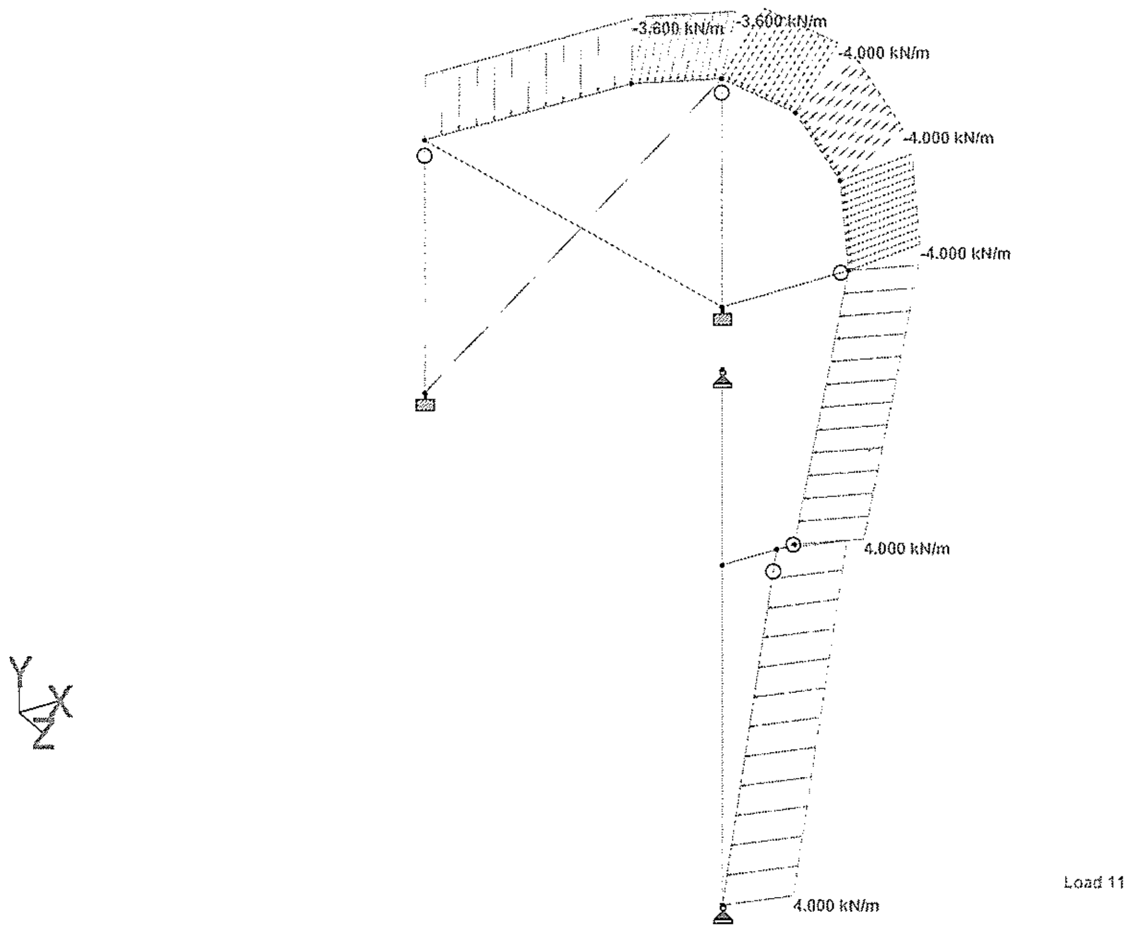
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



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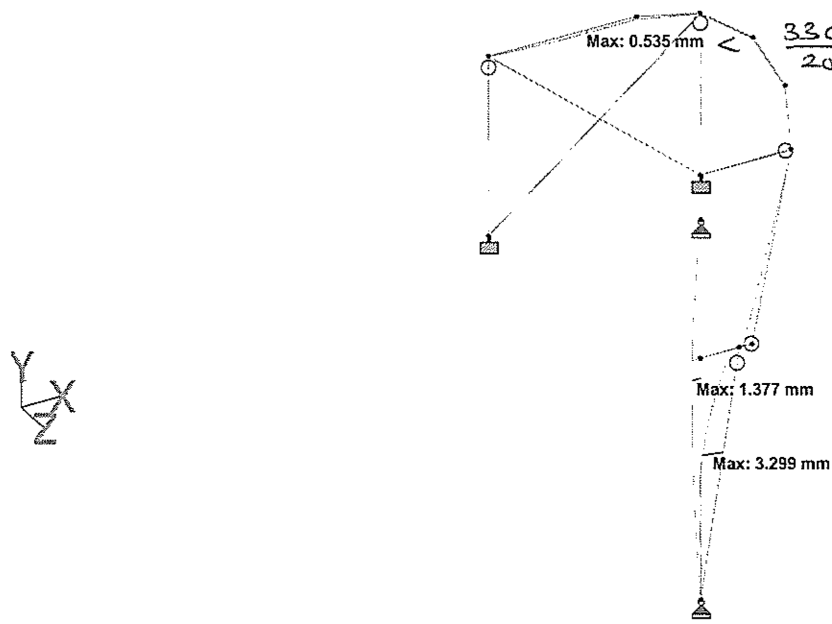
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



$< \frac{3300}{200} = 16.5$ mm Ok

$< \frac{6000}{200} = 30$ mm Ok

$< \frac{3800}{200} = 19$ mm Ok

Load 2011 : Displacement
Displacement - mm

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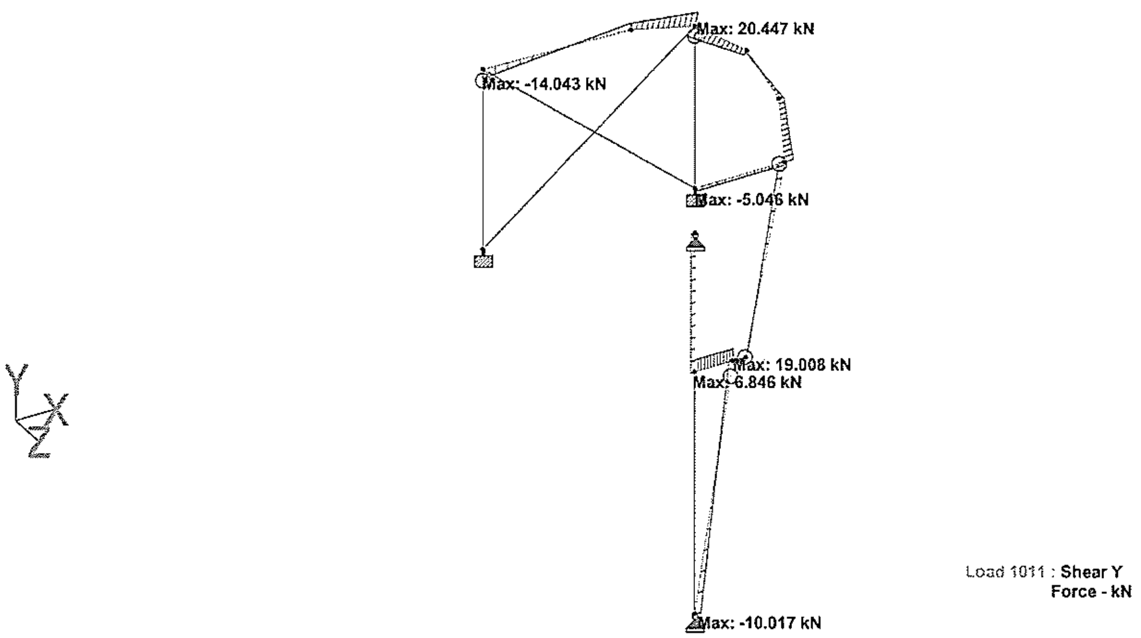
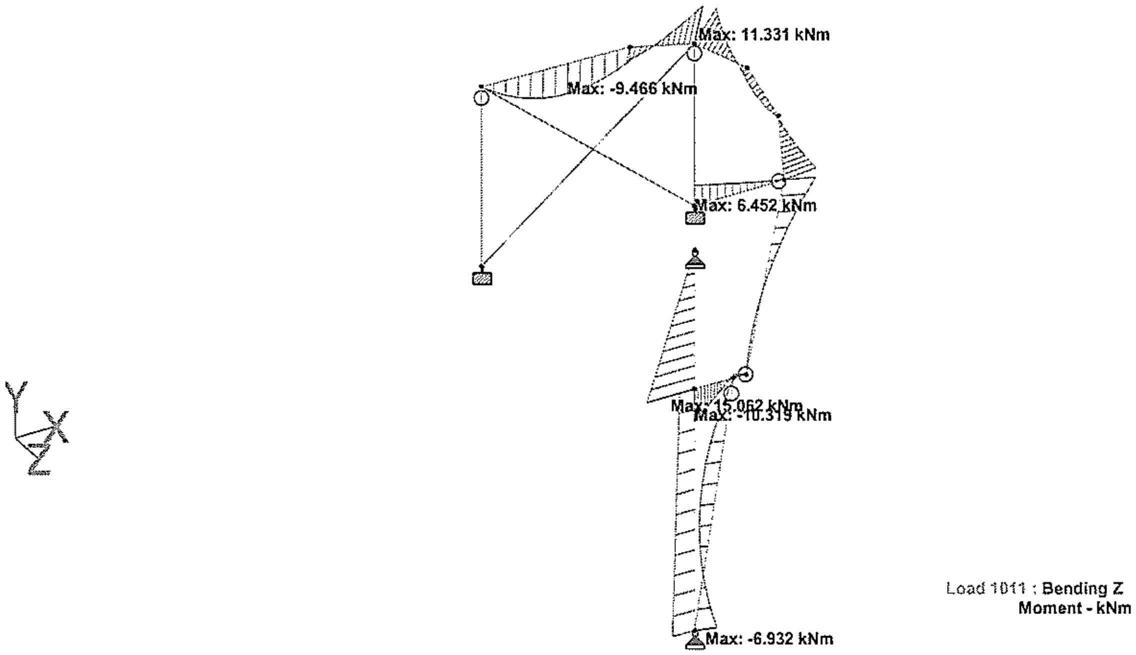
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



Software licensed to Meinhardt Infrastructure Pte Ltd	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54	



```
 ****************************************************
 *                                                  *
 *           STAAD.Pro                              *
 *           Version  2006     Bld 1005.US          *
 *           Proprietary Program of                 *
 *           Research Engineers,  Intl.             *
 *           Date=    JUL 18, 2007                  *
 *           Time=      15:59:43                    *
 *                                                  *
 *      USER ID: Meinhardt Infrastructure Pte Ltd   *
 ****************************************************

     1. STAAD PLANE
INPUT FILE: BRU Steel_Roof-TPS.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE JULY 2007
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1  19.656  3.800
     9. 2  20.273  6.705
    10. 3  20.183  7.739
    11. 4  19.679  8.647
    12. 5  18.850  9.272
    13. 6  17.839  9.506
    14. 7  15.501  9.547
    15. 11 18.850  6.705
    16. 12 15.501  6.705
    17. 51 19.456  3.800
    18. 52 18.850  3.800
    19. 61 18.850  0.000
    20. 62 18.850  6.000
    22. MEMBER INCIDENCES
    23. 1  1 2 6
    24. 11 2 11
    25. 12 11 5
    26. 13 12 7
    27. 21 12 5
    28. 22 7 11
    29. 51 1 51
    30. 52 51 52
    31. 61 61 51
    32. 71 61 52
    33. 72 52 62
    35. DEFINE MATERIAL START
    36. ISOTROPIC STEEL
    37. E 2.05E+008
    38. POISSON 0.3
    39. DENSITY 76.8195
    40. ALPHA 1.2E-005
    41. DAMP 0.03
    42. END DEFINE MATERIAL
```



```
    44. MEMBER PROPERTY BRITISH
    45. 1 TO 6 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    46. 11 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    47. 12 13 TABLE ST TUB1501506.3
    48. 21 22 TABLE ST UA90X90X6
    49. 51 52 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    50. 61 TABLE ST TUB1401406.0
    51. 71 72 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
    53. CONSTANTS
    54. MATERIAL STEEL ALL
    55. MEMBER TENSION
    56. 21 22
    58. MEMBER RELEASES
    59. 11 51 START MX MY MZ
    60. 12 13 61 END MX MY MZ
    62. SUPPORTS
    63. 61 62 PINNED
    64. 11 12 FIXED
    66. LOAD 1 SELFWEIGHT
    67. SELFWEIGHT Y -1
    69. LOAD 2 DEAD LOAD
    70. MEMBER LOAD
    71. *ASSUME ROOF M&E SERVICES LOAD = 0.3 KPA
    72. *ASSUME ROOF INSULATION AND ROOF SHEETING AND MEMBERS LOAD = 0.5 KPA
    73. 1 TO 6 61 UNI GY -2.56
```



```
    75. LOAD 3 LIVE LOAD
    76. *LL = 0.6 KPA
    77. MEMBER LOAD
    78. 3 TO 6 UNI GY -1.92
    80. LOAD 11 WL
    81. MEMBER LOAD
    82. 1 61 UNI Y 4
    83. 2 TO 4 UNI Y -4.0
    84. 5 6 UNI Y -3.6
    86. LOAD COMB 1001 1.4DL + 1.6LL
    87. 1 1.4 2 1.4 3 1.6
    88. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL
    89. 1 1.2 2 1.2 3 1.2 11 1.2
    91. LOAD COMB 2001 1.0DL + 1.0LL
    92. 1 1.0 2 1.0 3 1.0
    93. LOAD COMB 2011 DL + LL + WL
    94. 1 1.0 2 1.0 3 1.0 11 1.0
    96. PERFORM ANALYSIS
```



```
          P R O B L E M   S T A T I S T I C S
          -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    13/    16/     4
   ORIGINAL/FINAL BAND-WIDTH=     9/     3/     12 DOF
   TOTAL PRIMARY LOAD CASES =     4, TOTAL DEGREES OF FREEDOM =     29
   SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.0/ 359483.5 MB

 **START ITERATION NO.            2
 **START ITERATION NO.            3

 **NOTE-Tension/Compression converged after  3 iterations, Case=    1

 **START ITERATION NO.            2

 **NOTE-Tension/Compression converged after  2 iterations, Case=    2

 **START ITERATION NO.            2

 **NOTE-Tension/Compression converged after  2 iterations, Case=    3

 **START ITERATION NO.            2

 **NOTE-Tension/Compression converged after  2 iterations, Case=   11

    97. LOAD LIST ALL
    98. PRINT SUPPORT REACTION
 SUPPORT  REACTION
```



```
  SUPPORT REACTIONS -UNIT KN   METE    STRUCTURE TYPE = PLANE
  -----------------

 JOINT  LOAD   FORCE-X   FORCE-Y   FORCE-Z    MOM-X     MOM-Y     MOM Z

    61    1      0.47      5.05      0.00      0.00      0.00      0.00
          2      1.91     16.16      0.00      0.00      0.00      0.00
          3      0.23      1.16      0.00      0.00      0.00      0.00
         11      9.94      2.47      0.00      0.00      0.00      0.00
       1001      3.70     31.54      0.00      0.00      0.00      0.00
       1011     15.06     29.81      0.00      0.00      0.00      0.00
       2001      2.61     22.36      0.00      0.00      0.00      0.00
       2011     12.55     24.84      0.00      0.00      0.00      0.00
    62    1     -0.18      2.10      0.00      0.00      0.00      0.00
          2     -0.93      1.52      0.00      0.00      0.00      0.00
          3      0.00     -0.07      0.00      0.00      0.00      0.00
         11      6.82    -11.56      0.00      0.00      0.00      0.00
       1001     -1.56      4.96      0.00      0.00      0.00      0.00
       1011      6.85     -9.60      0.00      0.00      0.00      0.00
```

```
      2001     -1.11       3.56     0.00       0.00       0.00       0.00
      2011      5.71      -8.00     0.00       0.00       0.00       0.00
  11     1     -0.16       4.76     0.00       0.00       0.00       0.79
         2     -0.49      13.39     0.00       0.00       0.00       2.56
         3     -0.24       6.49     0.00       0.00       0.00       0.34
        11     21.30      16.35     0.00       0.00       0.00       1.73
      1001     -1.29      35.80     0.00       0.00       0.00       5.22
      1011     24.50      49.19     0.00       0.00       0.00       6.50
      2001     -0.89      24.64     0.00       0.00       0.00       3.68
      2011     20.42      41.00     0.00       0.00       0.00       5.42
  12     1     -0.12       1.83     0.00       0.00       0.00       0.01
         2     -0.49       2.99     0.00       0.00       0.00       0.03
         3      0.01       2.89     0.00       0.00       0.00      -0.02
        11      0.02       5.59     0.00       0.00       0.00      -0.05
      1001     -0.85      11.38     0.00       0.00       0.00       0.02
      1011     -0.71      15.96     0.00       0.00       0.00      -0.03
      2001     -0.61       7.71     0.00       0.00       0.00       0.02
      2011     -0.59      13.30     0.00       0.00       0.00      -0.03

************** END OF LATEST ANALYSIS RESULT **************

  99. PRINT FORCE ENVELOPE NSECTION 5 ALL
 FORCE    ENVELOPE NSECTION 5
    STAAD PLANE                                           -- PAGE NO.    5

        MEMBER FORCE ENVELOPE
        ---------------------

        ALL UNITS ARE KN    METE

MEMB  DISTANCE          FY    LD         MZ    LD         FZ    LD         MY    LD

   1    0.00  MAX      1.16 1001       0.00 2011       0.00 2011       0.00 2011
              MIN     -2.91   11       0.00 2011       0.00 2011       0.00 2011
        0.59  MAX      0.62 1001       1.02   11       0.00 2011       0.00 2011
              MIN     -0.53   11      -0.53 1001       0.00 2011       0.00 2011
        1.19  MAX      2.27 1011       0.63   11       0.00 2011       0.00 2011
              MIN      0.00    3      -0.73 1001       0.00 2011       0.00 2011
        1.78  MAX      4.65 1011       0.00    3       0.00 2011       0.00 2011
              MIN     -0.47 1001      -1.92 1011       0.00 2011       0.00 2011
        2.38  MAX      7.04 1011       0.00    3       0.00 2011       0.00 2011
              MIN     -1.02 1001      -5.40 1011       0.00 2011       0.00 2011
        2.97  MAX      9.42 1011       0.60 1001       0.00 2011       0.00 2011
              MIN     -1.57 1001     -10.28 1011       0.00 2011       0.00 2011

 ------------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB        1, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD        FX        DIST  LD

 MAX.       9.42    2.97 1011        1.02    0.59   11
            0.00    0.00    1        0.00    0.00    1     15.79 C    0.00 1001
 MIN.      -2.91    0.00   11      -10.28    2.97 1011
            0.00    2.97 2011        0.00    2.97 2011      6.19 T    2.97   11
 ------------------------------------------------------------------------------

   2    0.00  MAX     12.94 1011      10.28 1011       0.00 2011       0.00 2011
              MIN     -0.13    3      -0.60 1001       0.00 2011       0.00 2011
        0.21  MAX     11.87 1011       7.71 1011       0.00 2011       0.00 2011
              MIN     -0.13    3      -0.79 1001       0.00 2011       0.00 2011
        0.42  MAX     10.81 1011       5.36 1011       0.00 2011       0.00 2011
              MIN     -0.13    3      -0.97 1001       0.00 2011       0.00 2011
        0.62  MAX      9.74 1011       3.51   11       0.00 2011       0.00 2011
              MIN     -0.13    3      -1.13 1001       0.00 2011       0.00 2011
        0.83  MAX      8.68 1011       2.02   11       0.00 2011       0.00 2011
              MIN.    -0.13    3      -1.28 1001       0.00 2011       0.00 2011
        1.04  MAX      7.61 1011       0.71   11       0.00 2011       0.00 2011
              MIN     -0.13    3      -1.41 1001       0.00 2011       0.00 2011
    STAAD PLANE                                           -- PAGE NO.    6

 ------------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB        2, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD        FX        DIST  LD

 MAX.      12.94    0.00 1011       10.28    0.00 1011
```



```
            0.00    0.00    1        0.00    0.00    1      5.49 C    0.00 1001
 MIN.      -0.13    1.04    3       -1.41    1.04 1001
            0.00    1.04 2011        0.00    1.04 2011      4.06 T    1.04 1011
 ------------------------------------------------------------------------------

   3    0.00  MAX      5.30 1011       0.71   11       0.00 2011       0.00 2011
              MIN      0.03    1      -1.41 1001       0.00 2011       0.00 2011
        0.21  MAX      3.69 1011       0.07    3       0.00 2011       0.00 2011
              MIN     -0.13    2      -1.52 1001       0.00 2011       0.00 2011
        0.42  MAX      2.17   11       0.04    3       0.00 2011       0.00 2011
              MIN     -0.61 1001      -1.91 1011       0.00 2011       0.00 2011
        0.62  MAX      1.34   11       0.05    3       0.00 2011       0.00 2011
              MIN     -1.36 1001      -2.18 1011       0.00 2011       0.00 2011
        0.83  MAX      0.51   11       0.11    3       0.00 2011       0.00 2011
              MIN     -2.12 1001      -2.10 1011       0.00 2011       0.00 2011
        1.04  MAX     -0.27    1       0.20    3       0.00 2011       0.00 2011
              MIN     -2.87 1001      -1.70 1011       0.00 2011       0.00 2011

 ------------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB        3, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD        FX        DIST  LD

 MAX.       5.30    0.00 1011        0.71    0.00   11
            0.00    0.00    1        0.00    0.00    1      1.28 C    0.00    3
 MIN.      -2.87    1.04 1001       -2.18    0.62 1011
            0.00    1.04 2011        0.00    1.04 2011     12.34 T    1.04 1011
 ------------------------------------------------------------------------------

   4    0.00  MAX     -0.55    1       0.20    3       0.00 2011       0.00 2011
              MIN     -7.53 1011      -1.70 1011       0.00 2011       0.00 2011
        0.21  MAX     -0.65    1       0.81 1001       0.00 2011       0.00 2011
              MIN     -9.54 1011      -0.46   11       0.00 2011       0.00 2011
        0.42  MAX     -0.75    1       2.26 1001       0.00 2011       0.00 2011
              MIN    -11.55 1011       0.18    1       0.00 2011       0.00 2011
        0.62  MAX     -0.85    1       4.87 1011       0.00 2011       0.00 2011
              MIN    -13.55 1011       0.34    1       0.00 2011       0.00 2011
        0.83  MAX     -0.95    1       7.89 1011       0.00 2011       0.00 2011
              MIN    -15.56 1011       0.53    1       0.00 2011       0.00 2011
        1.04  MAX     -1.05    1      11.33 1011       0.00 2011       0.00 2011
              MIN    -17.57 1011       0.74    1       0.00 2011       0.00 2011
    STAAD PLANE                                           -- PAGE NO.    7

 ------------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB        4, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD        FX        DIST  LD

 MAX.      -0.55    0.00    1       11.33    1.04 1011
            0.00    0.00    1        0.00    0.00    1      0.20 T    0.00    3
 MIN.     -17.57    1.04 1011       -1.70    0.00 1011
            0.00    1.04 2011        0.00    1.04 2011     13.97 T    1.04 1011
 ------------------------------------------------------------------------------

   5    0.00  MAX     20.45 1011      11.33 1011       0.00 2011       0.00 2011
              MIN      1.21    1       0.74    1       0.00 2011       0.00 2011
        0.21  MAX     18.32 1011       7.31 1011       0.00 2011       0.00 2011
              MIN      1.08    1       0.50    1       0.00 2011       0.00 2011
        0.42  MAX     16.19 1011       3.73 1011       0.00 2011       0.00 2011
              MIN      0.96    1       0.28    1       0.00 2011       0.00 2011
        0.62  MAX     14.06 1011       0.59 1011       0.00 2011       0.00 2011
              MIN      0.84    1      -0.17    3       0.00 2011       0.00 2011
        0.83  MAX     11.93 1011      -0.06    1       0.00 2011       0.00 2011
              MIN      0.72    1      -2.11 1011       0.00 2011       0.00 2011
        1.04  MAX      9.80 1011      -0.20    1       0.00 2011       0.00 2011
              MIN      0.60    1      -4.36 1011       0.00 2011       0.00 2011

 ------------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB        5, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD        FX        DIST  LD

 MAX.      20.45    0.00 1011       11.33    0.00 1011
            0.00    0.00    1        0.00    0.00    1      4.90 C    0.00 1011
 MIN.       0.60    1.04    1       -4.36    1.04 1011
            0.00    1.04 2011        0.00    1.04 2011      0.14 C    1.04    1
 ------------------------------------------------------------------------------
```

```
 6     0.00 MAX     10.31 1011     -0.20    1      0.00 2011      0.00 2011
            MIN      0.61    1     -4.36 1011      0.00 2011      0.00 2011
       0.47 MAX      5.44 1011     -0.42    1      0.00 2011      0.00 2011
            MIN      0.33    1     -8.05 1011      0.00 2011      0.00 2011
       0.94 MAX      0.57 1011     -0.51    1      0.00 2011      0.00 2011
            MIN      0.00    3     -9.45 1011      0.00 2011      0.00 2011
       1.40 MAX     -0.23    1     -0.47    1      0.00 2011      0.00 2011
            MIN     -4.30 1011     -8.58 1011      0.00 2011      0.00 2011
       1.87 MAX     -0.51    1     -0.30    1      0.00 2011      0.00 2011
            MIN     -9.17 1011     -5.43 1011      0.00 2011      0.00 2011
       2.34 MAX     -0.79    1      0.00 2011      0.00 2011      0.00 2011
            MIN    -14.04 1011      0.00 2011      0.00 2011      0.00 2011
 STAAD PLANE                                               -- PAGE NO.    8

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB       6, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.       10.31   0.00 1011        0.00   2.34    1
 |             0.00   0.00    1        0.00   0.00    1       1.36 C     0.00 1011
 | MIN.      -14.04   2.34 1011       -9.45   0.94 1011
 |             0.00   2.34 2011        0.00   2.34 2011       0.07 T     2.34    2
 |-----------------------------------------------------------------------------

 11    0.00 MAX     -0.12    1      0.00 2011      0.00 2011      0.00 2011
            MIN     -4.02 1011      0.00 2011      0.00 2011      0.00 2011
       0.28 MAX     -0.24    3      1.17 1011      0.00 2011      0.00 2011
            MIN     -4.23 1011      0.06    1      0.00 2011      0.00 2011
       0.57 MAX     -0.24    3      2.41 1011      0.00 2011      0.00 2011
            MIN     -4.43 1011      0.14    3      0.00 2011      0.00 2011
       0.85 MAX     -0.24    3      3.70 1011      0.00 2011      0.00 2011
            MIN     -4.64 1011      0.20    3      0.00 2011      0.00 2011
       1.14 MAX     -0.24    3      5.04 1011      0.00 2011      0.00 2011
            MIN     -4.84 1011      0.27    3      0.00 2011      0.00 2011
       1.42 MAX     -0.24    3      6.45 1011      0.00 2011      0.00 2011
            MIN     -5.05 1011      0.34    3      0.00 2011      0.00 2011

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      11, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.       -0.12   0.00    1        6.45   1.42 1011
 |             0.00   0.00    1        0.00   0.00    1      23.17 C     0.00 1011
 | MIN.       -5.05   1.42 1011        0.00   0.00 2011
 |             0.00   1.42 2011        0.00   1.42 2011       1.64 T     1.42 1001
 |-----------------------------------------------------------------------------

 12    0.00 MAX      0.04 1001      0.09 1001      0.00 2011      0.00 2011
            MIN     -0.01   11     -0.03   11      0.00 2011      0.00 2011
       0.51 MAX      0.04 1001      0.07 1001      0.00 2011      0.00 2011
            MIN     -0.01   11     -0.02   11      0.00 2011      0.00 2011
       1.03 MAX      0.04 1001      0.05 1001      0.00 2011      0.00 2011
            MIN     -0.01   11     -0.02   11      0.00 2011      0.00 2011
       1.54 MAX      0.04 1001      0.04 1001      0.00 2011      0.00 2011
            MIN     -0.01   11     -0.01   11      0.00 2011      0.00 2011
       2.05 MAX      0.04 1001      0.02 1001      0.00 2011      0.00 2011
            MIN     -0.01   11     -0.01   11      0.00 2011      0.00 2011
       2.57 MAX      0.04 1001      0.00 2011      0.00 2011      0.00 2011
            MIN     -0.01   11      0.00 2011      0.00 2011      0.00 2011
 STAAD PLANE                                               -- PAGE NO.    9

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      12, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.        0.04   0.00 1001        0.09   0.00 1001
 |             0.00   0.00    1        0.00   0.00    1      45.07 C     0.00 1011
 | MIN.       -0.01   2.57   11       -0.03   0.00   11
 |             0.00   2.57 2011        0.00   2.57 2011       2.91 C     2.57    1
 |-----------------------------------------------------------------------------

 13    0.00 MAX      0.01    2      0.03    2      0.00 2011      0.00 2011
            MIN     -0.02   11     -0.05   11      0.00 2011      0.00 2011
       0.57 MAX      0.01    2      0.02    2      0.00 2011      0.00 2011
            MIN     -0.02   11     -0.04   11      0.00 2011      0.00 2011
```



```
       1.14 MAX      0.01    2      0.02    2      0.00 2011      0.00 2011
            MIN     -0.02   11     -0.03   11      0.00 2011      0.00 2011
       1.71 MAX      0.01    2      0.01    2      0.00 2011      0.00 2011
            MIN     -0.02   11     -0.02   11      0.00 2011      0.00 2011
       2.27 MAX      0.01    2      0.01    2      0.00 2011      0.00 2011
            MIN     -0.02   11     -0.01   11      0.00 2011      0.00 2011
       2.84 MAX      0.01    2      0.00 2011      0.00 2011      0.00 2011
            MIN     -0.02   11      0.00 2011      0.00 2011      0.00 2011

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      13, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.        0.01   0.00    2        0.03   0.00    2
 |             0.00   0.00    1        0.00   0.00    1      16.31 C     0.00 1011
 | MIN.       -0.02   2.84   11       -0.05   0.00   11
 |             0.00   2.84 2011        0.00   2.84 2011       0.97 C     2.84    1
 |-----------------------------------------------------------------------------

 21    0.00 MAX      0.19 1001      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00   11      0.00 2011      0.00 2011      0.00 2011
       0.84 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       1.69 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       2.53 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       3.38 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       4.22 MAX      0.00   11      0.00 2011      0.00 2011      0.00 2011
            MIN     -0.19 1001      0.00 2011      0.00 2011      0.00 2011
 STAAD PLANE                                               -- PAGE NO.   10

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      21, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.        0.19   0.00 1001        0.00   0.00    1
 |             0.00   0.00    1        0.00   0.00    1       0.00       0.00    3
 | MIN.       -0.19   4.22 1001        0.00   4.22 2011
 |             0.00   4.22 2011        0.00   4.22 2011       1.21 T     4.22 1001
 |-----------------------------------------------------------------------------

 22    0.00 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       0.88 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       1.76 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       2.64 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       3.51 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011
       4.39 MAX      0.00 2011      0.00 2011      0.00 2011      0.00 2011
            MIN      0.00 2011      0.00 2011      0.00 2011      0.00 2011

 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      22, AMONGST ALL SECT LOCATIONS
 |              FY/   DIST   LD        MZ/   DIST   LD
 |              FZ    DIST   LD        MY    DIST   LD        FX         DIST   LD
 |
 | MAX.        0.00   0.00    1        0.00   0.00    1
 |             0.00   0.00    1        0.00   0.00    1       0.00       0.00    1
 | MIN.        0.00   4.39 2011        0.00   4.39 2011
 |             0.00   4.39 2011        0.00   4.39 2011       1.76 T     4.39 1011
 |-----------------------------------------------------------------------------

 51    0.00 MAX      6.66   11      0.00 2011      0.00 2011      0.00 2011
            MIN    -15.68 1001      0.00 2011      0.00 2011      0.00 2011
       0.04 MAX      6.66   11      0.63 1001      0.00 2011      0.00 2011
            MIN    -15.72 1001     -0.27   11      0.00 2011      0.00 2011
       0.08 MAX      6.66   11      1.26 1001      0.00 2011      0.00 2011
            MIN    -15.75 1001     -0.53   11      0.00 2011      0.00 2011
       0.12 MAX      6.66   11      1.89 1001      0.00 2011      0.00 2011
            MIN    -15.78 1001     -0.80   11      0.00 2011      0.00 2011
       0.16 MAX      6.66   11      2.52 1001      0.00 2011      0.00 2011
            MIN    -15.82 1001     -1.07   11      0.00 2011      0.00 2011
```

```
      0.20  MAX      6.66   11       3.15 1001        0.00 2011        0.00 2011
            MIN    -15.85 1001      -1.33   11        0.00 2011        0.00 2011
    STAAD PLANE                                                 -- PAGE NO.   11


 -----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     51, AMONGST ALL SECT LOCATIONS
 |          FY/    DIST   LD          MZ/    DIST   LD
 |          FZ     DIST   LD          MY     DIST   LD        FX         DIST   LD
 |
 | MAX.     6.66   0.00   11          3.15   0.20 1001
 |          0.00   0.00    1          0.00   0.00    1       3.67 C     0.00 1011
 | MIN.   -15.85   0.20 1001         -1.33   0.20   11
 |          0.00   0.20 2011          0.00   0.20 2011       0.23 C     0.20    3
 -----------------------------------------------------------------------------


 52   0.00  MAX     19.01 1011       3.15 1001        0.00 2011        0.00 2011
            MIN     -3.35 1001      -1.33   11        0.00 2011        0.00 2011
      0.12  MAX     18.92 1011       3.57 1001        0.00 2011        0.00 2011
            MIN     -3.45 1001      -3.54   11        0.00 2011        0.00 2011
      0.24  MAX     18.83 1011       3.99 1001        0.00 2011        0.00 2011
            MIN     -3.55 1001      -5.76   11        0.00 2011        0.00 2011
      0.36  MAX     18.75 1011       4.43 1001        0.00 2011        0.00 2011
            MIN     -3.66 1001      -7.97   11        0.00 2011        0.00 2011
      0.48  MAX     18.66 1011       4.88 1001        0.00 2011        0.00 2011
            MIN     -3.76 1001     -10.18   11        0.00 2011        0.00 2011
      0.61  MAX     18.57 1011       5.34 1001        0.00 2011        0.00 2011
            MIN     -3.86 1001     -12.39   11        0.00 2011        0.00 2011


 -----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     52, AMONGST ALL SECT LOCATIONS
 |          FY/    DIST   LD          MZ/    DIST   LD
 |          FZ     DIST   LD          MY     DIST   LD        FX         DIST   LD
 |
 | MAX.    19.01   0.00 1011          5.34   0.61 1001
 |          0.00   0.00    1          0.00   0.00    1       6.27 C     0.00 1011
 | MIN.    -3.86   0.61 1001        -12.39   0.61   11
 |          0.00   0.61 2011          0.00   0.61 2011       0.76 T     0.61 1001
 -----------------------------------------------------------------------------


 61   0.00  MAX      1.48 1001       1.12 1001        0.00 2011        0.00 2011
            MIN    -10.02 1011      -6.93 1011        0.00 2011        0.00 2011
      0.77  MAX      1.01 1001       0.17 1001        0.00 2011        0.00 2011
            MIN     -6.73 1011      -0.52   11        0.00 2011        0.00 2011
      1.54  MAX      0.53 1001       3.43 1011        0.00 2011        0.00 2011
            MIN     -3.44 1011      -0.42 1001        0.00 2011        0.00 2011
      2.31  MAX      0.05 1001       4.81 1011        0.00 2011        0.00 2011
            MIN     -0.17   11      -0.65 1001        0.00 2011        0.00 2011
      3.08  MAX      3.13 1011       3.67 1011        0.00 2011        0.00 2011
            MIN     -0.42 1001      -0.51 1001        0.00 2011        0.00 2011
      3.85  MAX      6.41 1011       0.00 2011        0.00 2011        0.00 2011
            MIN     -0.90 1001       0.00 2011        0.00 2011        0.00 2011
    STAAD PLANE                                                 -- PAGE NO.   12


 -----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     61, AMONGST ALL SECT LOCATIONS
 |          FY/    DIST   LD          MZ/    DIST   LD
 |          FZ     DIST   LD          MY     DIST   LD        FX         DIST   LD
 |
 | MAX.     6.41   3.85 1011          4.81   2.31 1011
 |          0.00   0.00    1          0.00   0.00    1      36.49 C     0.00 1011
 | MIN.   -10.02   0.00 1011         -6.93   0.00 1011
 |          0.00   3.85 2011          0.00   3.85 2011       1.21 C     3.85    3
 -----------------------------------------------------------------------------


 71   0.00  MAX      1.04   11       6.93 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001      -1.12 1001        0.00 2011        0.00 2011
      0.76  MAX      1.04   11       6.49 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001      -0.52 1001        0.00 2011        0.00 2011
      1.52  MAX      1.04   11       6.06 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001      -0.04    2        0.00 2011        0.00 2011
      2.28  MAX      1.04   11       5.62 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001       0.09    3        0.00 2011        0.00 2011
      3.04  MAX      1.04   11       5.18 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001       0.12    3        0.00 2011        0.00 2011
      3.80  MAX      1.04   11       4.74 1011        0.00 2011        0.00 2011
            MIN     -0.80 1001       0.15    3        0.00 2011        0.00 2011
```



```
 -----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     71, AMONGST ALL SECT LOCATIONS
 |          FY/    DIST   LD          MZ/    DIST   LD
 |          FZ     DIST   LD          MY     DIST   LD        FX         DIST   LD
 |
 | MAX.     1.04   0.00   11          6.93   0.00 1011
 |          0.00   0.00    1          0.00   0.00    1       3.93 C     0.00 1001
 | MIN.    -0.80   3.80 1001         -1.12   0.00 1001
 |          0.00   3.80 2011          0.00   3.80 2011       7.39 T     3.80 1011
 -----------------------------------------------------------------------------


 72   0.00  MAX      6.85 1011      15.06 1011        0.00 2011        0.00 2011
            MIN     -1.56 1001      -3.43 1001        0.00 2011        0.00 2011
      0.44  MAX      6.85 1011      12.05 1011        0.00 2011        0.00 2011
            MIN     -1.56 1001      -2.74 1001        0.00 2011        0.00 2011
      0.88  MAX      6.85 1011       9.04 1011        0.00 2011        0.00 2011
            MIN     -1.56 1001      -2.06 1001        0.00 2011        0.00 2011
      1.32  MAX      6.85 1011       6.02 1011        0.00 2011        0.00 2011
            MIN     -1.56 1001      -1.37 1001        0.00 2011        0.00 2011
      1.76  MAX      6.85 1011       3.01 1011        0.00 2011        0.00 2011
            MIN     -1.56 1001      -0.69 1001        0.00 2011        0.00 2011
      2.20  MAX      6.85 1011       0.00 2011        0.00 2011        0.00 2011
            MIN     -1.56 1001       0.00 2011        0.00 2011        0.00 2011
    STAAD PLANE                                                 -- PAGE NO.   13


 -----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     72, AMONGST ALL SECT LOCATIONS
 |          FY/    DIST   LD          MZ/    DIST   LD
 |          FZ     DIST   LD          MY     DIST   LD        FX         DIST   LD
 |
 | MAX.     6.85   0.00 1011         15.06   0.00 1011
 |          0.00   0.00    1          0.00   0.00    1      11.56 C     0.00   11
 | MIN.    -1.56   2.20 1001         -3.43   0.00 1001
 |          0.00   2.20 2011          0.00   2.20 2011       4.96 T     2.20 1001
 -----------------------------------------------------------------------------

 ********** END OF FORCE ENVELOPE FROM INTERNAL STORAGE **********


 100. LOAD LIST 2001 2011
 101. PRINT JOINT DISPL
JOINT    DISPL
    STAAD PLANE                                                 -- PAGE NO.   14


 JOINT DISPLACEMENT (CM   RADIANS)    STRUCTURE TYPE = PLANE
 ------------------

 JOINT  LOAD   X-TRANS   Y-TRANS   Z-TRANS   X-ROTAN   Y-ROTAN   Z-ROTAN


     1 2001    0.0049   -0.0144    0.0000    0.0000    0.0000    0.0000
       2011   -0.1358    0.0066    0.0000    0.0000    0.0000   -0.0005
     2 2001    0.0001   -0.0146    0.0000    0.0000    0.0000    0.0000
       2011   -0.0017   -0.0220    0.0000    0.0000    0.0000   -0.0002
     3 2001   -0.0016   -0.0149    0.0000    0.0000    0.0000    0.0000
       2011   -0.0025   -0.0219    0.0000    0.0000    0.0000    0.0000
     4 2001    0.0023   -0.0126    0.0000    0.0000    0.0000   -0.0001
       2011   -0.0017   -0.0208    0.0000    0.0000    0.0000   -0.0001
     5 2001    0.0057   -0.0075    0.0000    0.0000    0.0000    0.0001
       2011    0.0034   -0.0130    0.0000    0.0000    0.0000    0.0001
     6 2001    0.0016   -0.0260    0.0000    0.0000    0.0000    0.0001
       2011   -0.0037   -0.0448    0.0000    0.0000    0.0000    0.0002
     7 2001    0.0020   -0.0029    0.0000    0.0000    0.0000   -0.0003
       2011   -0.0029   -0.0051    0.0000    0.0000    0.0000   -0.0005
    11 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
       2011    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    12 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
       2011    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    51 2001    0.0049   -0.0093    0.0000    0.0000    0.0000   -0.0002
       2011   -0.1358    0.0067    0.0000    0.0000    0.0000    0.0000
    52 2001    0.0049   -0.0004    0.0000    0.0000    0.0000   -0.0001
       2011   -0.1356    0.0012    0.0000    0.0000    0.0000   -0.0001
    61 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
       2011    0.0000    0.0000    0.0000    0.0000    0.0000    0.0009
    62 2001    0.0000    0.0000    0.0000    0.0000    0.0000    0.0001
       2011    0.0000    0.0000    0.0000    0.0000    0.0000   -0.0008

 ************** END OF LATEST ANALYSIS RESULT **************
```


033

```
 102. PARAMETER 1
 103. CODE BS5950
 104. PY 345000 ALL
 105. TRACK 2 ALL
 106. CHECK CODE ALL
STEEL DESIGN


                    STAAD.Pro CODE CHECKING - (BSI )
                    ***********************

 PROGRAM CODE REVISION V2.9_5950-1_2000
    STAAD PLANE                                        -- PAGE NO.   15


 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX           MY              MZ       LOCATION
 =======================================================================

      1   TAPERED       PASS     BS-4.8.2.2          0.033      2011
                         4.25 T        0.00           8.57          2.97
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     1 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2164.0   PT=  2693.8     MB=  267.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      2   TAPERED       PASS     BS-4.8.2.2          0.032      2011
                         0.11 T        0.00           8.57          0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     2 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.2   PT=  2693.8     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      3   TAPERED       PASS     BS-4.2.3-(Y)        0.011      2011
                         5.67 T        0.00          -0.32          0.00
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     3 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.1   PT=  2693.8     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      4   TAPERED       PASS     BS-4.8.2.2          0.039      2011
                        11.64 T        0.00          -9.44          1.04
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     4 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC= 11908.2   PT=  2693.8     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      5   TAPERED       PASS     BS-4.2.3-(Y)        0.041      2011
                         4.09 C        0.00           9.44          0.00
    STAAD PLANE                                        -- PAGE NO.   16


 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX           MY              MZ       LOCATION
 =======================================================================

 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     5 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2677.3   PT=     0.0     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
```



```
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
      6   TAPERED       PASS     BS-4.7   (C)        0.030      2011
                         1.04 C        0.00          -7.89          0.97
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB     6 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2346.9   PT=     0.0     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
     11   TAPERED       PASS     BS-4.7   (C)        0.028      2011
                        19.31 C        0.00          -5.38          1.42
 |-----------------------------------------------------------------------|
 | CALCULATED CAPACITIES FOR MEMB    11 UNIT - kN,m  SECTION CLASS  3    |
 | MCZ=  270.6   MCY=  111.6   PC=  2583.3   PT=  2693.8     MB=  270.6  |
 | BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00   |
 |  Shear Buckling check is required : Vb =  332 kN : qw      = 207 N/mm2 |
 |  d    = 200 mm  :    t  =   8 mm         : a  =    0 mm : pyf = 344 N/mm2 |
 |  BS-4.4.3.2      status = PASS          : BS-4.4.3.3 status = PASS     |
 |-----------------------------------------------------------------------|
    STAAD PLANE                                        -- PAGE NO.   17


 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                          FX           MY              MZ       LOCATION
 =======================================================================

     12 ST TUB1501506.3  PASS    BS-4.8.3.3.3        0.034      2011
                        37.56 C        0.00           0.04          -
 =======================================================================
   MATERIAL DATA
     Grade of steel             =  S 275
     Modulus of elasticity      =  205 kN/mm2
     Design Strength  (py)      =  344  N/mm2

   SECTION PROPERTIES (units - cm)
     Member Length =     256.70
     Gross Area =   35.80  Net Area =   35.80  Eff. Area =   35.80

                                     z-z  axis        y-y  axis
     Moment of inertia          :     1223.000         1223.000
     Plastic modulus            :      192.000          192.000
     Elastic modulus            :      163.067          163.067
     Effective modulus          :      163.067          163.067
     Shear Area                 :       17.900           17.900

   DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class              :   SEMI-COMPACT
     Squash Load                :  1235.10
     Axial force/Squash load    :    0.030

                                     z-z  axis        y-y  axis
     Compression Capacity       :      1141.3           1141.3
     Moment Capacity            :        66.2             66.2
     Reduced Moment Capacity    :        66.2             66.2
     Shear Capacity             :       370.5            370.5

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                                     x-x axis          y-y axis
     Slenderness                :      43.919           43.919
     Radius of gyration (cm)    :       5.845            5.845
     Effective Length           :       2.567            2.567

     LTB check unnecessary for this section

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

     CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
   BS-4.7    (C)    0.033  2011    37.6       -        -        -        -
   BS-4.8.3.2       0.030  2011    37.0      0.0      0.0      0.0      0.0
   BS-4.8.3.3.1     0.034  2011    37.6       -        -       0.0      0.0
   BS-4.8.3.3.3     0.034  2011    37.6       -        -       0.0      0.0
     Torsion and deflections have not been considered in the design.
                          ____________________
    STAAD PLANE                                        -- PAGE NO.   18
```


034

```
ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/      RATIO/     LOADING/
                           FX           MY              MZ        LOCATION
=======================================================================

    13 ST TUB1501506.3  PASS     BS-4.8.3.3.3       0.013      2011
                       13.59 C        0.00            0.03         -
=======================================================================
  MATERIAL DATA
    Grade of steel           =  S 275
    Modulus of elasticity    =  205 kN/mm2
    Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
    Member Length =     284.20
    Gross Area =   35.80  Net Area =    35.80  Eff. Area =    35.80

                                  z-z  axis        y-y  axis
    Moment of inertia        :      1223.000        1223.000
    Plastic modulus          :       192.000         192.000
    Elastic modulus          :       163.067         163.067
    Effective modulus        :       163.067         163.067
    Shear Area               :        17.900          17.900

  DESIGN DATA (units - kN,m)  BS5950-1/2000
    Section Class            :   SEMI-COMPACT
    Squash Load              :   1235.10
    Axial force/Squash load  :     0.011

                                  z-z  axis        y-y  axis
    Compression Capacity     :       1117.9          1117.9
    Moment Capacity          :         66.2            66.2
    Reduced Moment Capacity  :         66.2            66.2
    Shear Capacity           :        370.5           370.5

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)
                                  x-x axis         y-y axis
    Slenderness              :       48.624          48.624
    Radius of gyration (cm)  :        5.845           5.845
    Effective Length         :        2.842           2.842

    LTB check unnecessary for this section

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.7    (C)   0.012  2011     13.6       -        -        -        -
  BS-4.8.3.2      0.011  2011     13.1      0.0      0.0      0.0      0.0
  BS-4.8.3.3.1    0.013  2011     13.6       -        -       0.0      0.0
  BS-4.8.3.3.3    0.013  2011     13.6       -        -       0.0      0.0
   Torsion and deflections have not been considered in the design.

     STAAD PLANE                                         -- PAGE NO.    19

ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/      RATIO/     LOADING/
                           FX           MY              MZ        LOCATION
=======================================================================

    21 ST UA90X90X6       PASS     BS-4.6    (T)      0.002      2001
                        0.87 T        0.00            0.00         4.22
=======================================================================
  MATERIAL DATA
    Grade of steel           =  S 275
    Modulus of elasticity    =  205 kN/mm2
    Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
    Member Length =     421.96
    Gross Area =   10.60  Net Area =    10.60  Eff. Area =    10.60

                                  z-z  axis        y-y  axis
    Moment of inertia        :        33.585         131.618
    Plastic modulus          :        17.554          37.227
    Elastic modulus          :         9.752          20.682
    Effective modulus        :         9.752          20.682
    Shear Area               :         3.240           3.240
```



```
  DESIGN DATA (units - kN,m)  BS5950-1/2000
    Section Class            :   SLENDER

                                  z-z  axis        y-y  axis
    Tension Capacity         :        365.7           365.7
    Shear Capacity           :         67.1            67.1

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

    LTB Moment Capacity (kNm) and LTB Length (m):     4.09,     4.220
    LTB Coefficients & Associated Moments (kNm):
    mLT =   1.00  :   mx =   1.00   :  my =   1.00   :  myx =   1.00
    Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.6    (T)   0.002  2001     0.9        -        -        -        -
   Torsion and deflections have not been considered in the design.

     STAAD PLANE                                         -- PAGE NO.    20

ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/      RATIO/     LOADING/
                           FX           MY              MZ        LOCATION
=======================================================================

    22 ST UA90X90X6       PASS     BS-4.6    (T)      0.004      2011
                        1.47 T        0.00            0.00         0.00
=======================================================================
  MATERIAL DATA
    Grade of steel           =  S 275
    Modulus of elasticity    =  205 kN/mm2
    Design Strength  (py)    =  344  N/mm2

  SECTION PROPERTIES (units - cm)
    Member Length =     439.24
    Gross Area =   10.60  Net Area =    10.60  Eff. Area =    10.60

                                  z-z  axis        y-y  axis
    Moment of inertia        :        33.585         131.618
    Plastic modulus          :        17.554          37.227
    Elastic modulus          :         9.752          20.682
    Effective modulus        :         9.752          20.682
    Shear Area               :         3.240           3.240

  DESIGN DATA (units - kN,m)  BS5950-1/2000
    Section Class            :   SLENDER

                                  z-z  axis        y-y  axis
    Tension Capacity         :        365.7           365.7
    Shear Capacity           :         67.1            67.1

  BUCKLING CALCULATIONS (units - kN,m)
  (axis nomenclature as per design code)

    LTB Moment Capacity (kNm) and LTB Length (m):     3.99,     4.392
    LTB Coefficients & Associated Moments (kNm):
    mLT =   1.00  :   mx =   1.00   :  my =   1.00   :  myx =   1.00
    Mlt =    0.00 :   Mx =    0.00  :  My =   0.00   :   My =   0.00

  CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

   CLAUSE         RATIO  LOAD     FX        VY       VZ       MZ       MY
  BS-4.6    (T)   0.004  2011     1.5        -        -        -        -
   Torsion and deflections have not been considered in the design.

    51     TAPERED        PASS     BS-4.2.3-(Y)       0.027      2001
                        1.50 C        0.00           -2.22         0.20
     STAAD PLANE                                         -- PAGE NO.    21

ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE        RESULT/   CRITICAL COND/      RATIO/     LOADING/
                           FX           MY              MZ        LOCATION
=======================================================================
```

```
|-----------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB    51 UNIT - kN,m  SECTION CLASS  3 |
| MCZ=  270.6    MCY=  111.6   PC=  2693.8   PT=     0.0    MB=  270.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 1.00 |
|  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2 |
|  d    = 200 mm  :   t  =    8 mm          : a  =    0 mm : pyf = 344 N/mm2 |
|  BS-4.4.3.2      status = PASS           : BS-4.4.3.3 status = PASS   |
|-----------------------------------------------------------------|
     52    TAPERED          PASS      BS-4.2.3-(Y)       0.038       2011
                           5.23 C           0.00          0.89         0.00
|-----------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB    52 UNIT - kN,m  SECTION CLASS  3 |
| MCZ=  270.6    MCY=  111.6   PC=  2693.8   PT=  2693.8    MB=  270.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 1.00 |
|  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2 |
|  d    = 200 mm  :   t  =    8 mm          : a  =    0 mm : pyf = 344 N/mm2 |
|  BS-4.4.3.2      status = PASS           : BS-4.4.3.3 status = PASS   |
|-----------------------------------------------------------------|
     STAAD PLANE                                          -- PAGE NO.   22


 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER      TABLE       RESULT/   CRITICAL COND/      RATIO/      LOADING/
                            FX            MY              MZ        LOCATION
 =======================================================================

     61 ST TUB1401406.0  PASS      ANNEX I.1          0.111       2011
                          30.41 C         0.00          5.78           -
 =======================================================================
   MATERIAL DATA
     Grade of steel           =  S 275
     Modulus of elasticity    =  205 kN/mm2
     Design Strength  (py)    =  344  N/mm2

   SECTION PROPERTIES (units - cm)
     Member Length =     384.80
     Gross Area =    31.80  Net Area =    31.80  Eff. Area =    31.80

                                  z-z  axis         y-y  axis
     Moment of inertia        :      944.000          944.000
     Plastic modulus          :      159.000          159.000
     Elastic modulus          :      134.857          134.857
     Effective modulus        :      134.857          134.857
     Shear Area               :       15.900           15.900

   DESIGN DATA (units - kN,m)   BS5950-1/2000
     Section Class            :   PLASTIC
     Squash Load              :   1097.10
     Axial force/Squash load  :     0.028

                                  z-z  axis         y-y  axis
     Compression Capacity     :      835.1            835.1
     Moment Capacity          :       54.9             54.9
     Reduced Moment Capacity  :       54.8             54.8
     Shear Capacity           :      329.1            329.1

   BUCKLING CALCULATIONS (units - kN,m)
   (axis nomenclature as per design code)
                                  x-x axis          y-y axis
     Slenderness              :      70.626           70.626
     Radius of gyration (cm)  :       5.448            5.448
     Effective Length         :       3.848            3.848

     LTB check unnecessary for this section

   CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

    CLAUSE       RATIO  LOAD     FX        VY        VZ        MZ        MY
   BS-4.2.3-(Y)  0.025  2011      -        8.3        -         -         -
   BS-4.7   (C)  0.036  2011    30.4        -         -         -         -
   BS-4.8.3.2    0.024  2011    30.4       8.3       0.0       5.8       0.0
   BS-4.8.3.3.1  0.161  2011    30.4        -         -        5.8       0.0
   BS-4.8.3.3.3  0.144  2011    30.4        -         -        5.8       0.0
   ANNEX I.1     0.111  2011    30.4        -         -        5.8       0.0
     STAAD PLANE                                          -- PAGE NO.   23

    Torsion and deflections have not been considered in the design.

     71    TAPERED          PASS      BS-4.8.2.2         0.023       2011
                           3.88 T           0.00          5.78         0.00
```



```
|-----------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB    71 UNIT - kN,m  SECTION CLASS  3 |
| MCZ=  270.6    MCY=  111.6   PC=  1928.4   PT=  2693.8    MB=  221.7 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 1.00 |
|  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2 |
|  d    = 200 mm  :   t  =    8 mm          : a  =    0 mm : pyf = 344 N/mm2 |
|  BS-4.4.3.2      status = PASS           : BS-4.4.3.3 status = PASS   |
|-----------------------------------------------------------------|
     72    TAPERED          PASS      BS-4.7   (C)       0.050       2011
                           9.32 C           0.00         12.55         0.00
|-----------------------------------------------------------------|
| CALCULATED CAPACITIES FOR MEMB    72 UNIT - kN,m  SECTION CLASS  3 |
| MCZ=  270.6    MCY=  111.6   PC=  2384.5   PT=  2693.8    MB=  270.6 |
| BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 1.00, my = 1.00, myx = 1.00 |
|  Shear Buckling check is required : Vb =  332 kN : qw       = 207 N/mm2 |
|  d    = 200 mm  :   t  =    8 mm          : a  =    0 mm : pyf = 344 N/mm2 |
|  BS-4.4.3.2      status = PASS           : BS-4.4.3.3 status = PASS   |
|-----------------------------------------------------------------|


 ************** END OF TABULATED RESULT OF DESIGN **************

 107. FINISH

          *********** END OF THE STAAD.Pro RUN ***********

             **** DATE= JUL 18,2007   TIME= 15:59:44 ****
```

3 ROOF BEAMS AND SIDE BEAM DESIGN

Site	Dubai Depot - Jebel Ali		Date	Jul '07	Job No.	068010
	TPS (Building 9)					
	Roof Secondary Beam Design		Designed by	Rudy	Sheet No.	

DL				
Selfweight	0.40 x	1.00 =	0.4	
Sheeting	0.10 x	3.80 =	0.38	
Purlins	0.10 x	3.80 =	0.38	
Insulation	0.30 x	3.80 =	1.14	
M&E Finishes	0.30 x	3.80 =	1.14	
			3.44	kN/m
LL	0.60 x	3.80 =	2.28	kN/m

w(sls) = 5.72 kN/m
w(uls) = 8.5 kN/m

a) For simply supported case with L = 3.6 m
Mmax = 13.7 kNm
Vmax = 15.2 kN

b) For cantilever case with L = 3.2 m
Mmax = 43.3 kNm
Vmax = 27.1 kN

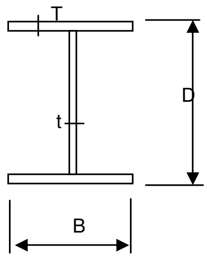
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai Depot - Jebel Ali	Date	Jul '07	Job No.	068010
	TPS (Building 9)				
	Roof Secondary Beam Design	Design	Rudy	Sheet No.	

Steel Beam - **Check for Mb**
- **UB with equal flange**
- **For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections**

Section Properties **Build-up Section**



Section	=	**w**	("r" for Rolled section, "w" for Welded section)	
p_y	=	**355**	N/mm2	
E	=	**205000**	N/mm2	
D	=	250	mm	$b/T = 9.0 < 13\varepsilon$
B	=	150	mm	$d/t = 39.0 < 80\varepsilon$
t	=	6	mm	⟹ **Class 3 semi-compact section**
T	=	8	mm	**B1** *(Beam / Column Mark)*

A	=	38.04	cm2	=	3804	mm2
Ix	=	4155.77	cm4	=	41557652	mm4
Iy	=	450.45	cm4	=	4504500	mm4
Zx	=	332.46	cm3	=	332461	mm3
Zy	=	60.06	cm3	=	60060	mm3
Sx	=	372.53	cm3	=	372534	mm3
Sy	=	15.02	cm3	=	15015	mm3
rx	=	10.45	cm	=	104.5214069	mm
ry	=	3.44	cm	=	34.41145543	mm

Design Data

Mx	=	**43.3**	kNm		
V	=	**27.1**	kN		
L	=	**3.2**	m	Overall span	
Le	=	**4.32**	m	Effective length for Mb	(1.6*2.7)

Design for Bending Moment

λ	=	126		

From BS5950 clause. 4.3.6.7

u	=	1.00		
x	=	31.25		
λ/x	=	4.02		
v	=	0.86	Equal flange	1 for class 1 plastic
Sect'n class	=	**3.00**		2 for class 2 compact
βw	=	0.9		3 for class 3 semi-compact
				4 for slender

λ_{LT} = $uv\lambda(\beta_w)^{0.5}$ = **102**

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	**97**	N/mm2
Mb	=	32.2	kNm

From Table 18

m_{LT}	=	**0.6**							
Mb/m_{LT}	=	**53.7**	**kNm**	>	Mx	=	43.3	kNm	**OK !**

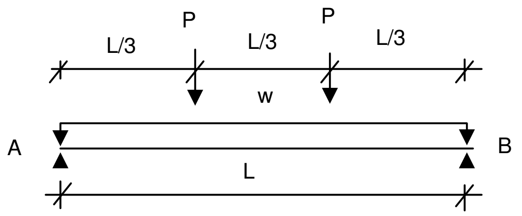
Design for Bending Shear

Av	=	1404	mm2						
Pv	=	299	kN						
0.6Pv	=	179	kN	>	V	=	27.1	kN	**Low shear, OK !**

Deflection Check

LL	=	**2.3**	kN/m	Uniform load		
M_{LL}	=	**11.8**	kNm			
d	=	1.4	mm	<	25mm	**OK!**
d/L	=	1/2261		<	1/250	**OK!**

Site	Dubai Depot - Rashidiya	Date	Jul '07	Job No.	068010
	TPS (Building 12)				
	Roof Beam Design (Above BRU Enclosure)	Designed by	Rudy	Sheet No.	



From STAAD Pro Analysis

P =	42.2	kN	(ULS)
	38.0	kN	(SLS)
w-selfwt =	0.8	kN/m	(ULS)
	0.6	kN/m	(SLS)
L =	9.6	m	

Reaction Ra = Rb = 46.2 kN (ULS)
40.9 kN

At x = L/3, M = 143.6 kNm (ULS)

Using 300 x 250 UB with T = 12 mm
t = 6 mm

E = 205000 N/mm2
Ix = 13500.03 cm4 = 135000288 mm4

Deflection due to point load
d1 = 9.4 mm

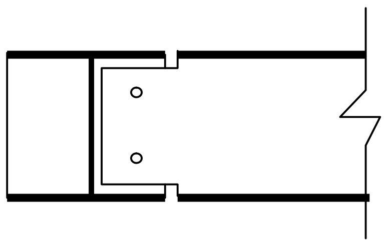
Max deflection due to selfwt =
d2 = 2.4 mm

Therefore max deflection = 11.8 mm **Ok, less than 25 or L/200**
(L/200 = 48 mm)

Bolt Connection Design

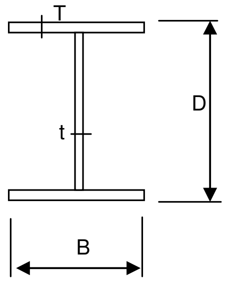
Shear, Fy	=	46.2	kN
Moment, Mz	=	0	kNm
Axial Force, Fx	=	0	kN
No of bolt provided	=	2	
Shear per bolt	=	23	kN
Bolt diameter	=	16	mm
Bolt's shear capacity	=	58.9	kN

Ok



Steel Beam — **- Check for Mb**
- UB with equal flange
- For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties — **Build-up Section**



Section	=	**w**	("r" for Rolled section, "w" for Welded section)			
p_y	=	**355**	N/mm2			
E	=	**205000**	N/mm2			
D	=	300	mm			
B	=	250	mm			
t	=	6	mm			
T	=	12	mm			

b/T = 10.2 < 13ε
d/t = 46.0 < 80ε
⟹ **Class 3 semi-compact section**

B1 *(Beam / Column Mark)*

A	=	76.56	cm2	=	7656	mm2
Ix	=	13500.03	cm4	=	135000288	mm4
Iy	=	3125.54	cm4	=	31255400	mm4
Zx	=	900.00	cm3	=	900002	mm3
Zy	=	250.04	cm3	=	250043	mm3
Sx	=	978.26	cm3	=	978264	mm3
Sy	=	62.51	cm3	=	62510.8	mm3
rx	=	13.28	cm	=	132.7903101	mm
ry	=	6.39	cm	=	63.89421933	mm

Design Data

Mx	=	**142.2**	kNm	
V	=	**46.2**	kN	
L	=	**9.6**	m	Overall span
Le	=	**3.2**	m	Effective length for Mb

Design for Bending Moment

λ	=	50	

From BS5950 clause. 4.3.6.7

u	=	1.00		
x	=	25.00		
λ/x	=	2.00		
v	=	0.96	Equal flange	1 for class 1 plastic
Sect'n class	=	**3.00**		2 for class 2 compact
βw	=	0.9		3 for class 3 semi-compact
				4 for slender

λ_{LT} = $u v \lambda (\beta_w)^{0.5}$ = **46**

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	**270**	N/mm2
Mb	=	243.0	kNm

From Table 18

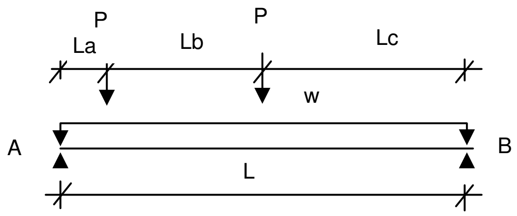
m_{LT}	=	**0.6**							
Mb/m_{LT}	=	**405.0**	**kNm**	>	Mx	=	142.2	kNm	**OK !**

Design for Bending Shear

Av	=	1656	mm2						
Pv	=	353	kN						
0.6Pv	=	212	kN	>	V	=	46.2	kN	**Low shear, OK !**

Calculation/Sketc **Meinhardt Infrastructure Pte Ltd**
Consulting Engineers, Planners & Managers

Site	Dubai Depot - Rashidiya	Date	Jul '07	Job No.	068010
	TPS (Building 12)				
	Roof Beam Design (Above BRU Enclosure)	Designed by	Rudy	Sheet No.	



From STAAD Pro Analysis

P =	46.2 kN	(ULS)
	38.0 kN	(SLS)
w-selfwt =	0.8 kN/m	(ULS)
	0.6 kN/m	(SLS)
L =	6.4 m	
La =	0.6 m	
Lb =	2.6 m	
Lc =	3.2 m	

Reaction	Ra =	67.7 kN	(ULS)
	Rb =	30.1 kN	(ULS)

at x = L/2, M = 174.6 kNm

Using 300 x 250 UB with T = 12 mm
t = 6 mm

E	=	205000	N/mm2		
Ix	=	13500.03	cm4	=	135000288 mm4

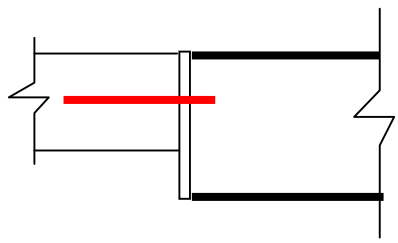
Deflection due to point load
d1 = 9.6 mm

Max deflection due to selfwt =
d2 = 0.5 mm

Therefore max deflection = 10.1 mm **Ok, less than 25 or L/200**
(L/200 = 32 mm)

Bolt Connection Design

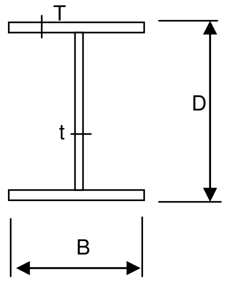
Shear, Fy	=	67.7 kN
Moment, Mz	=	0 kNm
Axial Force, Fx	=	0 kN
No of bolt provided	=	2
Shear per bolt	=	34 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN **Ok**



Site	Dubai Depot - Rashidiya	Date	Jul '07	Job No.	068010
	TPS (Building 12)				
	Roof Beam Design (Above BRU Enclosure)	Design	Rudy	Sheet No.	

Steel Beam
- **Check for Mb**
- **UB with equal flange**
- **For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections**

Section Properties **Build-up Section**



Section	=	**w**	("r" for Rolled section, "w" for Welded section)	
p_y	=	**355**	N/mm2	
E	=	**205000**	N/mm2	
D	=	300	mm	b/T = 10.2 < 13ε
B	=	250	mm	d/t = 46.0 < 80ε
t	=	6	mm	⇒ **Class 3 semi-compact section**
T	=	12	mm	**B1** *(Beam / Column Mark)*

A	=	76.56	cm2	=	7656	mm2
Ix	=	13500.03	cm4	=	135000288	mm4
Iy	=	3125.54	cm4	=	31255400	mm4
Zx	=	900.00	cm3	=	900002	mm3
Zy	=	250.04	cm3	=	250043	mm3
Sx	=	978.26	cm3	=	978264	mm3
Sy	=	62.51	cm3	=	62510.8	mm3
rx	=	13.28	cm	=	132.7903101	mm
ry	=	6.39	cm	=	63.89421933	mm

Design Data

Mx	=	**174.6**	kNm	
V	=	**67.7**	kN	
L	=	**6.4**	m	Overall span
Le	=	**3.2**	m	Effective length for Mb

Design for Bending Moment

λ = 50

From BS5950 clause. 4.3.6.7

u	=	1.00		
x	=	25.00		
λ/x	=	2.00		
v	=	0.96	Equal flange	1 for class 1 plastic
Sect'n class	=	**3.00**		2 for class 2 compact
βw	=	0.9		3 for class 3 semi-compact
				4 for slender

λ_{LT} = $uv\lambda(\beta_w)^{0.5}$ = **46**

From Table 16 (rolled section) or Table 17 (welded section)

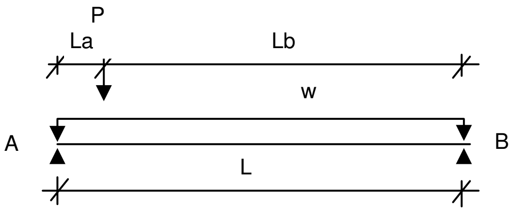
p_b	=	**270**	N/mm2
Mb	=	243.0	kNm

From Table 18

m_{LT}	=	**0.6**							
Mb/m_{LT}	=	**405.0**	**kNm**	>	Mx	=	174.6	kNm	**OK !**

Design for Bending Shear

Av	=	1656	mm2						
Pv	=	353	kN						
0.6Pv	=	212	kN	>	V	=	67.7	kN	**Low shear, OK !**



From STAAD Pro Analysis

P =	46.2	kN	(ULS)
	38.0	kN	(SLS)
w-selfwt =	0.8	kN/m	(ULS)
	0.6	kN/m	(SLS)
L =	3.2	m	
La =	0.6	m	
Lb =	2.6	m	

Reaction	Ra =	38.9	kN	(ULS)
	Rb =	8.5	kN	(ULS)
at x = La,	M =	23.2	kNm	

Using 250 x 250 UB with T = 8 mm
t = 6 mm

E	=	205000	N/mm2		
Ix	=	6499.18	cm4	=	64991785.3 mm4

Deflection due to point load
d1 = 3.0 mm

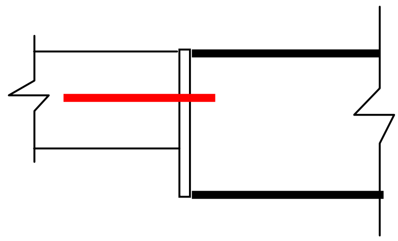
Max deflection due to selfwt =
d2 = 0.1 mm

Therefore max deflection = 3.1 mm **Ok, less than 25 or L/200**
(L/200 = 16 mm)

Bolt Connection Design

Shear, Fy	=	38.9	kN
Moment, Mz	=	0	kNm
Axial Force, Fx	=	0	kN
No of bolt provided	=	2	
Shear per bolt	=	19	kN
Bolt diameter	=	16	mm
Bolt's shear capacity	=	58.9	kN

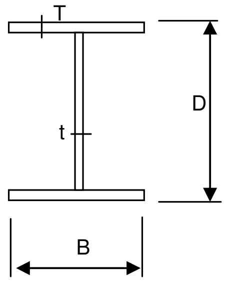
Ok



Site	Dubai Depot - Rashidiya	Date	Jul '07	Job No.	068010
	TPS (Building 12)				
	Roof Beam Design (Above BRU Enclosure)	Design	Rudy	Sheet No.	

Steel Beam - **Check for Mb**
- **UB with equal flange**
- **For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections**

Section Properties **Build-up Section**



Section	=	**w**	("r" for Rolled section, "w" for Welded section)			
p_y	=	**355**	N/mm2			
E	=	**205000**	N/mm2			
D	=	250	mm		b/T = 15.3 > 13ε	
B	=	250	mm		d/t = 39.0 < 80ε	
t	=	6	mm		**Slender section !**	
T	=	8	mm		**B1** *(Beam / Column Mark)*	
A	=	54.04	cm2	=	5404	mm2
Ix	=	6499.18	cm4	=	64991785.33	mm4
Iy	=	2083.78	cm4	=	20837833.33	mm4
Zx	=	519.93	cm3	=	519934	mm3
Zy	=	166.70	cm3	=	166703	mm3
Sx	=	566.13	cm3	=	566134	mm3
Sy	=	41.68	cm3	=	41675.66667	mm3
rx	=	10.97	cm	=	109.665889	mm
ry	=	6.21	cm	=	62.09671269	mm

Design Data

Mx	=	**23.2**	kNm	
V	=	**38.9**	kN	
L	=	**3.2**	m	Overall span
Le	=	**3.2**	m	Effective length for Mb

Design for Bending Moment

λ = 52

From BS5950 clause. 4.3.6.7

u	=	1.00		
x	=	31.25		
λ/x	=	1.65		
v	=	0.97	Equal flange	1 for class 1 plastic
Sect'n class	=	**3.00**		2 for class 2 compact
βw	=	0.9		3 for class 3 semi-compact
				4 for slender

λ_{LT} = $u v \lambda (\beta_w)^{0.5}$ = **48**

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	**253**	N/mm2
Mb	=	131.5	kNm

From Table 18

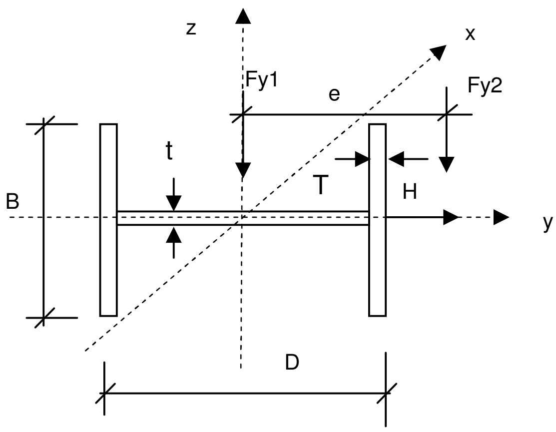
m_{LT}	=	**0.6**							
Mb/m_{LT}	=	**219.2**	**kNm**	>	Mx	=	23.2	kNm	**OK !**

Design for Bending Shear

Av	=	1404	mm2						
Pv	=	299	kN						
0.6Pv	=	179	kN	>	V	=	38.9	kN	**Low shear, OK !**

Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Rashidiya - Bldg 12)				
	Side Beam Design	Designed by	Rudy	Sheet No.	

Member Properties



Beam depth, D	=	250	mm
Beam width, B	=	250	mm
Flange thickness, T	=	15	mm
Web thickness, t	=	8	mm
Depth btw fillets, d	=	220	mm
Sectional area, Ag	=	9260	mm^2
Moment of inertia, Iz	=	1.11E+08	mm^4
Moment of inertia, Iy	=	3.91E+07	mm^4
Section Modulus, Zz	=	8.86E+05	mm^3
Section Modulus, Zy	=	3.13E+05	mm^3
* Plastic modulus, Sz	=	9.91E+05	mm^3
* Plastic modulus, Sy	=	3.32E+04	mm^3

[*calculated based on contribution from flanges & web]

Radius of gyration, rz	=	109.38	mm
Radius of gyration, ry	=	64.96	mm
A,eff	=	9260	mm^2
Zz,eff	=	8.30E+05	mm^3
Zy,eff	=	3.13E+05	mm^3

E	=	205000	N/mm^2
v	=	0.3	
G	=	E / 2*(1+v)	
	=	78846	N/mm^2
Steel grade	=	50	
Strength, py	=	355	N/mm^2

Section Classification

Epsilon = [275/py]^0.5	=	0.880
Outstand element of compression flange, b/T		
Plastic	=	7.9
Compact	=	8.8
Semi-compact	=	13.2
Actual b/T	=	8.066667

Flange is compact.

Web subject to compression throughout, d/t

Axial Force P_{max}	=	0.00	kN
r_1	=	0.00	
Plastic	=	70.4	
Compact	=	88.0	
r_2	=	0.00	
Semi-compact	=	105.62	
Actual d/t	=	27.50	

Web is plastic.

Section classified as **Compact**

For symmetrical I & H section

Torsional constant, J	=	$(1/3)*[2BT^3 + (D-2T)*t^3]$	=	6.00E+05
Warping constant, H	=	$0.25*Iy*(D-T)^2$	=	5.394E+11
Warping static moment, Sws	=	$(D-T)*B^2T / 16$	=	1.38E+07
Normalised warping function, Wns	=	(D-T)B/4	=	1.47E+04

Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Rashidiya - Bldg 12)				
	Side Beam Design	Designed by	Rudy	Sheet No.	

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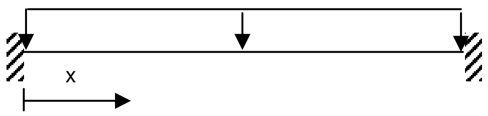

Span Length, Lx = 3.2 m
$Mx = 0.5wL_x^2 [(x/L_x) - (x/L_x)^2 - 1/6]$
e, eccentricity = 350 mm

Fy1 - Self wt = 0.73 kN/m

Fy2 - Purlin + Cladding + Insulation

Height	=	4.5	m
Weight	=	0.5	kN/m^2
Load	=	2.3	kN/m

H - Wind Load

Pressure	=	1.24	kN/m^2
Load	=	5.6	kN/m

For bending along minor axis (w = 1.4DL)

At x = 0 and x = L	My(max)	=	$wLx^2/12$	=	3.6 kNm
At x = 0.25L and x =0.75L	My	=	$wL^2/96$	=	0.4 kNm
At x =0.5L	My	=	$wL^2/24$	=	1.8 kNm

m_{LT} = 0.2+ (0.15M2+0.5M3+0.15M4) / Mmax = 0.49
m_{LT} * Mymax = 1.7 kNm

For Bending along major axis (w = H)

At x = 0 and x = L	Mz(max)	=	$wLx^2/12$	=	4.8 kNm
At x = 0.25L and x =0.75L	Mz	=	$wLx^2/96$	=	
At x =0.5L	Mz	=	$wLx^2/24$	=	

m_{LT} = 0.2+ (0.15M2+0.5M3+0.15M4) / Mmax = 0.20
m_{LT} * Mzmax = 1.0 kNm

Torsional moment (w = 1.4Fy2)	Tq (max)	=	w*e*Lx	=	3.5 kNm
Shear in z-direction (w = 1.4DL)	Fvx	=	wLx/2	=	6.7 kN
Shear in y-direction (w = H)	Fvy	=	wLx/2	=	8.9 kN
Resultant shear	Fr	=	$(Fvx^2+Fvy^2)^{0.5}$	=	11.1 kN

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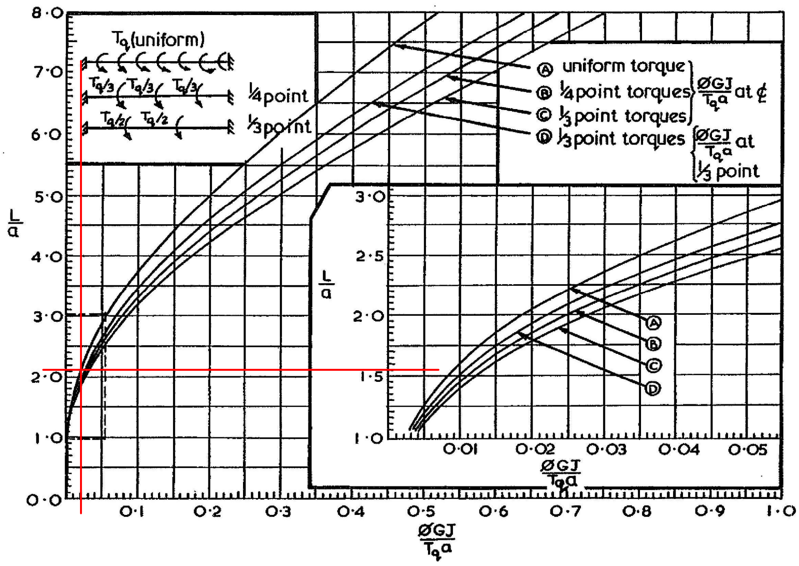
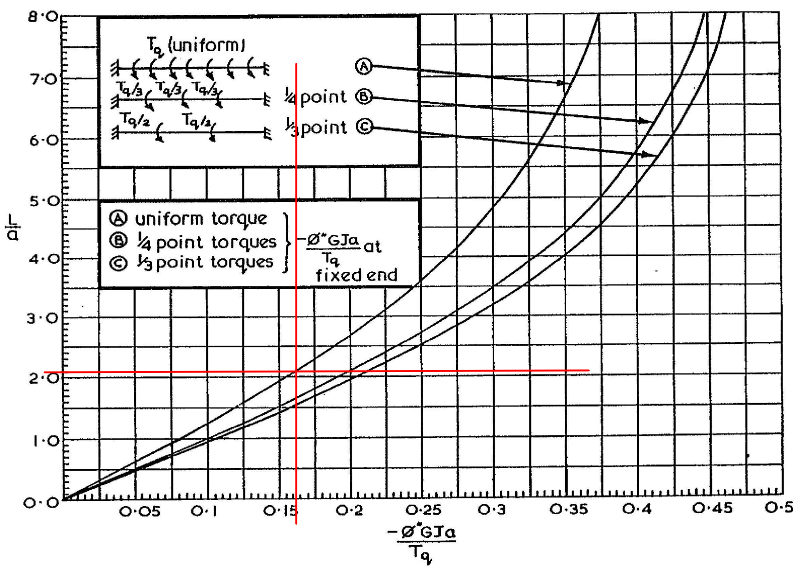
GRAPH 7



GRAPH 8



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Shear Capacity Check

Shear capacity of beam , P_v = $0.6p_yA_v$ = 426 kN **> Design Value, OK**

Since d / t is not more than 63 ε

Shear buckling resistance check is not required *Clause 4.2.3*

Local capacity check

Torsional bending constant, a = $(EH/GJ)^{0.5}$ = 1528.8477 mm

With uniform torque and L / a = 2.1

From the Graph 7

ϕ = 0.02 Tq*a/GJ

= 0.002 rad

Mzt = ϕMz

= 0.0 kNm

From the Graph 8

ϕ'' = 0.165 Tq/GJ*a

= 8.05E-09

σ_w = E*Wns*ϕ''

= 24.2 N/mm^2

σ_{byt} = Myt / Zy

= 0.03 N/mm^2

σ_{by} = My / Zy

= 11.38 N/mm^2

σ_{bz} = Mz / Zz

= 5.37 N/mm^2

$\sigma_{bz} + \sigma_{by} + \sigma_{byt} + \sigma_w$ = 41.0 N/mm^2 Ok, less than py

Overall buckling check

$(A,eff/Ag)^{0.5}$ = 1.00

Effective length, Lzz = 4.4 m | Effective length, Lyy 4.4 m

Slenderness Ratio: = 40.6 | Slenderness Ratio: = 68.35

Design Lamda, λ = 68.4

β_w = 1.00

Torsional index, x = 16.7

λ/x = 4.1

For equal flanges

Buckling parameter, u = 0.9

Slenderness factor, ν = $1/ [(1 + 0.05 (\lambda/x)^2]^{0.25}$ = 0.86

Equivalent Slenderness Ratio λ_{LT} = $u\, \nu\, \lambda\, (\beta w)^{0.5}$ = 52.8

Limiting Slenderness Ratio λ_{L0} = $0.4(\pi^2E /py)^{0.5}$ = 17

α_{LT} = 7 (Appendix B - Cl 2.2)

η_{LT} = $\alpha_{LT}(\lambda_{LT} - \lambda_{L0})/1000$ = 0.250

Euler strength, pe = $(\pi^2 E / \lambda_{LT}^2)$ = 725.4 N/mm^2

ϕ_{LT} = $0.5*[py + (\eta_{LT} + 1)pe]$ = 631.0 N/mm^2

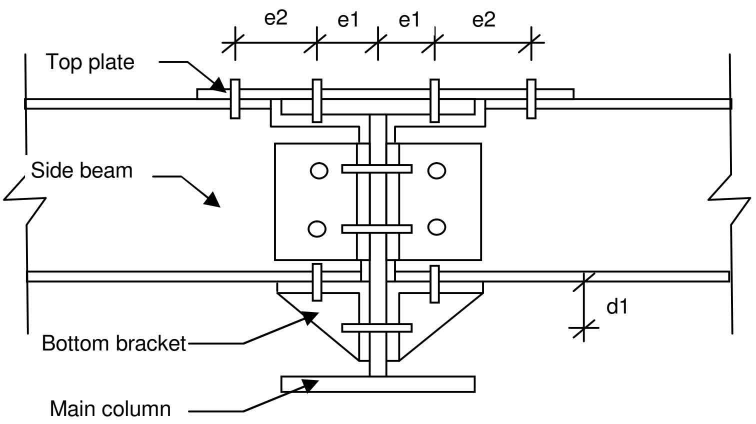
Bending strength, pbz = $pe*py / [\phi_{LT} + (\phi_{LT}^2 - pepy)^{0.5}]$ = 256.0 N/mm^2

Buckling Resis. Momt. Mbz = **253.6** kNm

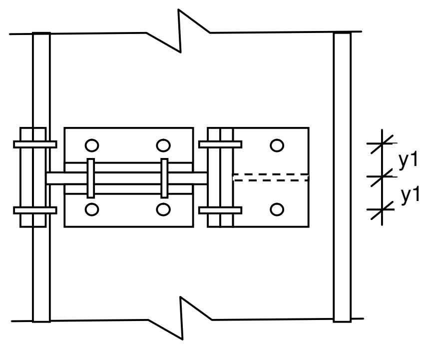
$m_{LT}Mz$ / Mbz + $m_{LT}My$ / pyZy + (σbyt + σw) / py * [1+ 0.5*(m_{LT}My/Mbz]

= 0.07 Ok, less than 1

Side Beam End Connection Design



Plan



Section

From the side beam design

Vertical bending moment	My =	3.6 kNm
Horizontal bending moment	Mz =	4.8 kNm
Torsional bending moment	Tq =	3.5 kNm
Vertical shear	Fy =	8.9 kN
Horizontal shear	Fz =	6.7 kN

Assumption
1) Vertical bending moment to be resisted by top and bottom bolt connected to main column
2) Horizontal and torsional bending moment to be resisted by bolt connected to side beam's flange
3) Vertical shear to be resisted by bolt on the side beam's flange and main column's web
4) Horizontal shear to be resisted by bolt on side beam's and main column's web

Side beam dimension

Depth	D1	=	250 mm
Width	B1	=	250 mm
Web thickness	t1	=	15 mm
Flange thickness	T1	=	8 mm

Main column dimension

Depth	D2	=	250 mm
Width	B2	=	250 mm
Web thickness	t2	=	12 mm
Flange thickness	T2	=	8 mm

e1	=	75 mm
e2	=	100 mm
d1	=	0 mm
y1	=	75 mm

Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Rashidiya - Bldg 12)				
	Side Beam Design	Designed by	Rudy	Sheet No.	

1) Bolts at the top and bottom of main column's web

a) Tension due to vertical bending moment

T = My / (2*y1*no of row of bolt) = 11.9 kN   no of row of bolt = 2

b) Shear due to vertical load

Vy = Fy / no of bolt = 2.2 kN   no of bolt = 4

c) Shear due to horizontal load

Vz = Fz / no of bolt = 1.7 kN   no of bolt = 4

d) Resultant shear

Vr = $[Vy^2 + Vz^2]^{0.5}$ = 2.8 kN

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN
Bolt tensile capacity, Tb = 70.7 kN

T / Tb + Vr / Vb = 0.21 Ok, not greater than 1.4

2) Bolts at the side beam's flange

a) Shear due to horizontal bending moment

Vhs = Mz / (D1*no of row of bolt) = 9.5 kN   no of row of bolt = 2

b) Shear due to torsional bending moment

Vt = Tq / (D1*no of bolt) = 7.1 kN   no of bolt = 2

c) Shear due to vertical load

Vy = Fy / no of bolt = 4.5 kN   no of bolt = 2

d) Resultant shear

Vr = $[Vhs^2 + (Vt + Vy)^2]^{0.5}$ = 14.9 kN

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN **Ok**

3) Bolt at the side beam's web

a) Shear due to horizontal load

Vz = Fz / no of bolt = 3.3 kN   no of bolt = 2

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN **Ok**

4) Bolts at the bracket connected to main column's web

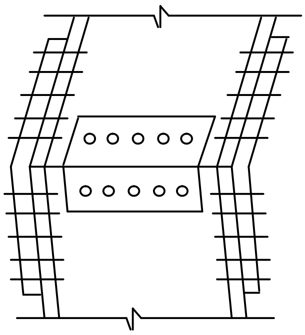
a) Tension due to horizontal bending moment

T = Mz / [(D1+d1)*no of row of bolt] = 9.5 kN   no of row of bolt = 2

Bolt diameter, ϕ = 16 mm
Bolt tensile capacity, Tb = 70.7 kN **Ok**

4 CONNECTION DESIGN

Column/Beam Splice Connection Design



From the STAAD Pro analysis

Moment, Mz	=	15.6 kNm
Shear, Fy	=	14.5 kN
Axial Force, Fx	=	37.8 kN
Beam - depth (D)	=	250 mm
width (B)	=	250 mm
web thickness (tw)	=	8 mm
flange thickness (tf)	=	12 mm

Assume that the direct horizontal and vertical shear are resisted by bolt on the beam web

Resultant shear, Vs	=	40 kN
No of row of bolt provided	=	1
No of bolt provided	=	2
Shear per bolt	=	20 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

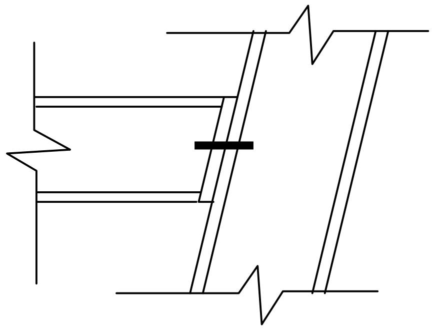
Assume shear due to bending is resisted by bolt on the beam flanges

Shear on bolt on the top and bottom flange due to Moment

Eccentricity of vertical load, e	=	50 mm
Moment due to eccentricity, Me	=	Fy*e
		0.7 kNm
Total moment, Mt	=	Me + Mz
	=	65.6 kNm
Shear on the bolt at top and bottom flan(	=	Mt / (D-tf)
	=	276 kN
No of row of bolt provided	=	2
No of bolt provided	=	2
Shear per bolt	=	69 kN
Bolt diameter	=	20 mm
Bolt's shear capacity	=	91.9 kN

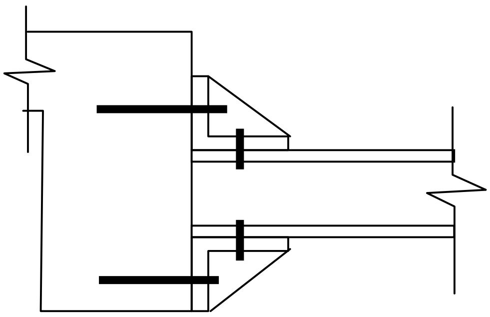
Cantilever Beam Connection Design

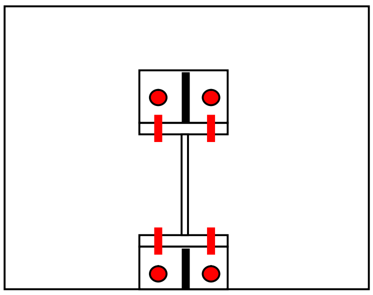
From the STAAD Pro analysis (Beam 11)
1) At the location connecting to side frame



Shear, Fy	=	1.42 kN		
No of row of bolt provided	=	1		
No of bolt provided	=	2		
Shear per bolt	=	1 kN		
Bolt diameter	=	16 mm		
Bolt's shear capacity	=	58.9 kN	**Ok**	
Weld size, s	=	4 mm		
Weld strength, pw	=	0.700 kN/mm		
Weld length required	=	Fy/pw		
	=	2	mm	
Weld length provided	=	1000	mm	**OK**

2) At the location connecting to roof beam / slab





From the STAAD Pro analysis (Beam 11)

Shear, Fy	=	2.37 kN
Moment, Mz	=	2.52 kNm
Axial Force, Fx	=	31.7 kN

a) Design of bolt at the beam top and bottom flange

Depth of beam, D	=	250 mm	
Flange thickness, T	=	12 mm	
Shear on the bolts, V	=	Mz/(D-T) + Fx	
	=	42.3 kN	
No of bolt provided	=	2	
Shear per bolt	=	21 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	**Ok**

b) Design of bolt connecting to RC Roof Slab / Beam

Distance between bolt, db	=	300 mm	
Tension force on the bolt, T	=	Mz/Db + Fx	
	=	40.1 kN	
No of bolt provided	=	2	
Tension force per bolt, T1	=	20 kN	
Bolt diameter	=	16 mm	
Bolt's tension capacity, Tb	=	70.3 kN	
Shear force per bolt, V1	=	Fy / no of bolt	
	=	1.2 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity, Vb	=	58.9 kN	
V1 / Vb + T1 / Tb	=	0.31	**Ok, not greater than 1**

c) Check weld connecting the stiffener plate to the angle

Weld size, s	=	4 mm	
Weld strength, pw	=	0.700 kN/mm	
Weld length required	=	max[(T+Fy),V]/pw	
	=	60.7 mm	
Weld length provided	=	284 mm	**OK**

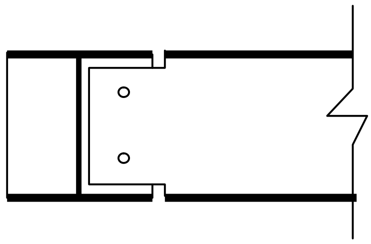
Calculation/Sketc

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai Depot	Date	Jul '07	Job No.	068010
	TPS (Rashidiya Bldg 12)				
	Side Frame to Roof Beam Connection	Designed by	Rudy	Sheet No.	

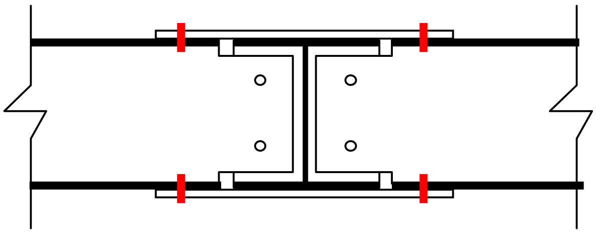
1) At the end of the frame

From the STAAD Pro analysis (beam 6)



Shear, Fy	=	14.8 kN	
Moment, Mz	=	0 kNm	
Axial Force, Fx	=	1.74 kN	
No of bolt provided	=	2	
Shear per bolt	=	7 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	**Ok**

2) At the end column support



From the STAAD Pro analysis (beam 6)

Shear, Fy	=	19.8 kN
Moment, Mz	=	9.0 kNm
Axial Load, Fx	=	4.7 kN

Assume the moment is resisted by bolts at the top and bottom flange

Beam depth, D	=	250 mm	
Flange thickness, T	=	8 mm	
Shear on the bolt, Vs	=	M/(D-T)	
	=	37.2	
No of row of bolt	=	1	
No of bolt provided	=	2	
Shear per bolt	=	19 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	**Ok**

Assume direct shear and axial load is resisted by bolts at the beam web

No of bolt provided	=	2	
Shear per bolt	=	10 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	**Ok**

Site	Dubai Depot	Date	Jul '07	Job No.	068010
	TPS (Rashidiya Bldg 12)				
	Beam Splice Connection	Designed by	Rudy	Sheet No.	

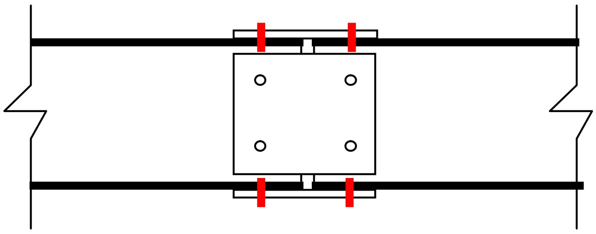
DL

Selfweight	0.40 x	1.00 =	0.4	
Sheeting	0.10 x	3.80 =	0.38	
Purlins	0.10 x	3.80 =	0.38	
Insulation	0.30 x	3.80 =	1.14	
M&E Finishes	0.30 x	3.80 =	1.14	
			3.44	kN/m
LL	0.60 x	3.80 =	2.28	kN/m

w(uls) = 8.5 kN/m

span L = 3.6 m
Mmax = 13.7 kNm
Vmax = 15.2 kN

Beam splice



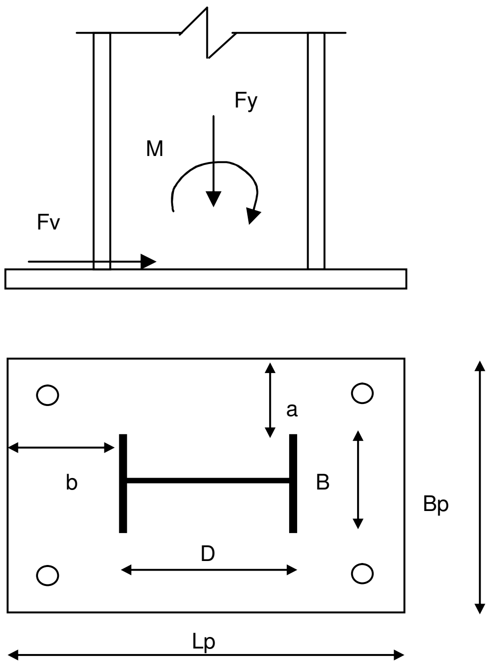
Assume the moment is resisted by bolts at the top and bottom flange

Beam depth, D	=	250	mm	
Flange thickness, T	=	8	mm	
Shear on the bolt, Vs	=	M/(D-T)		
	=	56.7		
No of row of bolt	=	1		
No of bolt provided	=	2		
Shear per bolt	=	28	kN	
Bolt diameter	=	16	mm	
Bolt's shear capacity	=	58.9	kN	**Ok**

Assume direct shear is resisted by bolts at the beam web

No of bolt provided	=	2		
Shear per bolt	=	8	kN	
Bolt diameter	=	16	mm	
Bolt's shear capacity	=	58.9	kN	**Ok**

Base Plate Design



<u>Applied Load</u>

Fy	=	37.7 kN
Fv	=	15.21 kN

<u>Column section</u>

Depth, D	=	250 mm
Width, B	=	250 mm
Flange thickness, tf	=	12 mm
Web thickness, tw	=	8 mm

<u>Base plate</u>

Length, Lp	=	350 mm
Width, Bp	=	350 mm
Thickness, tp	=	20 mm
pyp	=	345 N/mm2
fcu	=	40 N/mm^2
0.4 fcu	=	16 N/mm^2

Base plate area required = 2356.25 mm^2
Base plate area provided = 122500 mm^2
Base pressure, w = 0.30776 N/mm^2

a = 50 mm   b = 50 mm

Min. plate thickness required = $[2.5*w*(a^2-0.3b^2)/pyp]^{0.5}$
= 2.0 mm **Ok**

<u>Holding down bolt design</u>

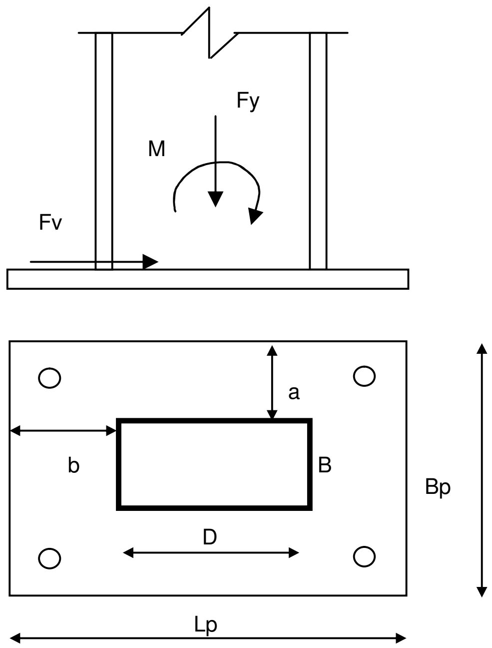
No of bolt provided	=	4
Maximum tension per bolt due to axial load, T1	=	0.0 kN
Shear per bolt, V1	=	3.8 kN

<u>Column to base plate weld design</u>

Weld size, s	=	6 mm		
Weld strength, pw	=	1.050 kN/mm		
Weld length required	=	(Fy+Fv)/pw		
	=	50	mm	
Weld length provided	=	1000	mm	**OK**

Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Rashidiya Bldg 12)				
	Roof Level	Designed by	Rudy	Sheet No.	

Base Plate Design



Applied Load

Fy	=	42.18 kN
Fv	=	32.94 kN
M	=	2.48 kNm

Column section

Depth, D	=	150 mm
Width, B	=	150 mm
Flange thickness, tf	=	6 mm
Web thickness, tw	=	6 mm

Base plate

Length, Lp	=	350 mm
Width, Bp	=	350 mm
Thickness, tp	=	20 mm
pyp	=	345 N/mm2
fcu	=	30 N/mm2
0.4 fcu	=	12 N/mm2

Holding down bolt design

Distance between the outer bolts, db	=	350 mm
No of bolt provided at the outer line	=	2
Total no of bolt	=	4
Maximum tension per bolt due to bending and axial load, T1	=	45.7 kN
Shear per bolt, V1	=	8.2 kN

Using grade 8.8 bolt,

Bolt size	=	16 mm
Tension capacity, Tb	=	70.7 kN
Shear capacity, Vb	=	58.9 kN
T1 / Tb + V1 / Vb	=	0.79 < = 1.4, Ok

Column to base plate weld design

Assuming moment is resisted by weld at the column flange

Force on the weld, T2	=	M / (Dc-tf)
	=	17.2 kN
Effective length of the weld resisting T2	=	2*B
	=	300 mm

Weld size, s =	4 mm	(up to 15mm thick)
Weld strength, pw =	0.7 kN/mm	(based on longitudinal capacity)

Total weld resistance	=	210.0 kN	Ok

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

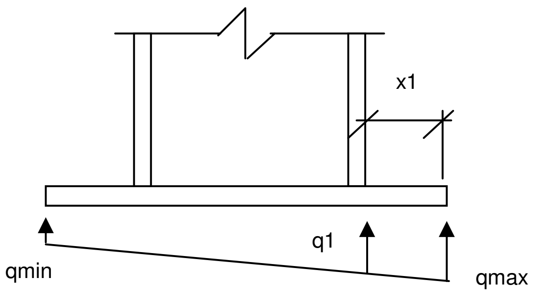
Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Rashidiya Bldg 12)				
	Roof Level	Designed by	Rudy	Sheet No.	

Assuming shear is resisted by weld at the web
Effective length of the weld resisting Fv = 300 mm

Weld size, s = 4 mm (up to 15mm thick)
Weld strength, pw = 0.7 kN/mm (based on longitudinal capacity)

Total weld resistance = 210 kN Ok

Design plate thickness



Distance between the outer bolt to the column edge, x1 = 50 mm
Maximum moment at the face of column, M1
= force on the bolt * distance = 4.6 kNm

Minimum thickness of plate required, tp min = $[\,6*M1 / 1.2\ pyp\,]^{0.5}$
= 8.1 mm Ok